

Card column reference

All **711** columns across the five cards, grouped by card and then by block, with blocks ordered by **measured importance**. Each entry leads with what the column *is*; the caveats fold underneath.

Generated 2026-08-20 by `analyses/ops/gen_column_reference.py` from `card_glossary.tsv` and `card_registry.tsv`. **Do not hand-edit**.

Vintage mix “ read before quoting a fitted column

The delivered edge card is **16 days newer** than the base it layers over. Its **84 base-owned columns** (`coupling_`, `beta_`, `dose_`, `share_`, `rank_`) carry the older vintage; the other 102 are current.

The one rule behind this layout

The rung is a property of (CARD, COLUMN) “ never of the column name. `beta` is edge-rung on the edge card and family-rung on the `gene_family` card: same estimator, different unit. That is why this is organised card-first, and why one name can appear twice with different text.

How blocks are ordered

By three measured signals, not by size: how often a block's columns are named in the **discovery registry** (what the findings rest on, weighted highest), how many **modules** read them, and whether the block is **base-owned** (fitted) rather than an annotation join. Names under 6 characters count only backtick-quoted registry hits “ otherwise `n` and `z` match ordinary prose.

One correction the measurement needed: mention-frequency is not importance “ a *settled* block gets discussed less precisely because nobody argues about it. `adm_` scored 27th of 44 on the raw signals while MH-216 calls `adm_has_site` the single most load-bearing conditioning variable, on 4 registry mentions. Five blocks therefore carry a **curated floor**, each citing the finding that justifies it (see `CURATED_FLOOR`). That is the only hand-set input here.

Cards

- edge “ 186 columns, 45 blocks
- gene “ 135 columns, 29 blocks
- gene_family “ 61 columns, 12 blocks
- arm “ 296 columns, 43 blocks
- seed_family “ 33 columns, 2 blocks

edge

One row per one (miRNA arm, gene) pair. The widest card and the one most claims are made from. 84 of its columns come from the BASE card and 102 from the annotation layer – check the vintage banner before quoting a fitted one.

186 columns 5,649 rows 45 blocks median coverage 72%

Contents – 45 blocks, most important first

1. (bare) 11 – CORE

2. coupling_ 13 – CORE

3. seed_ 1 – CORE

4. arm_ 15 – CORE

5. adm_ 4 – CORE

6. beta_ 8 – CORE

7. n_ 5 – CORE

8. share_ 4 – CORE

9. cal_ 7 – CORE

10. echim_ 3 – CORE

11. shift_ 1 – CORE

12. rank_ 4 – CORE

13. healthy_ 3 – SUPPORTING

14. ctx_ 18 – SUPPORTING

15. identity_ 5 – SUPPORTING

16. dose_ 7 – SUPPORTING

17. d_ 4 – SUPPORTING

18. gene_ 6 – SUPPORTING

19. ago_ 2 – SUPPORTING

20. pip_ 2 – SUPPORTING

21. retention_ 3 – REFERENCE

22. oof_ 3 – REFERENCE
23. grank_ 3 – REFERENCE

24. own_ 1 – REFERENCE

25. composition_ 1 – REFERENCE

26. w_ 1 – REFERENCE

27. m_ 1 – REFERENCE

28. prior_ 1 – REFERENCE

29. edge_ 1 – REFERENCE

30. term_ 3 – REFERENCE

31. mean_ 2 – REFERENCE

32. surrogate_ 2 – REFERENCE

33. esub_ 8 – REFERENCE

34. realization_ 2 – REFERENCE

35. wiring_ 1 – REFERENCE

36. kd_ 2 – REFERENCE

37. cell_ 1 – REFERENCE

38. sd_ 1 – REFERENCE

39. acquisition_ 1 – REFERENCE

40. z_ 1 – REFERENCE

41. heur_ 1 – REFERENCE

42. family_ 1 – REFERENCE

43. cptac_ 20 – REFERENCE

44. hly_ 1 – REFERENCE

45. is_ 1 – REFERENCE

(BARE) – 11

core – the columns findings rest on

Change in the arm's GLOBAL (cohort-wide) dose rank between two states. HLY_NAT is the FIELD delta, NAT_TUM the malignant step.

11 columns – coverage 63% – 100% (median 96%) – 3 share the block description
4× real, signed – 2× identifier – 2× real – 0 – 2× signed – 1 – 1 – 1× boolean

arm

KEY IDENTIFIER 100%

KEY – mature miRNA arm (e.g. hsa-miR-21-5p). Normalised through `_arm_key()`, which gates against the canonical universe and LOGS drops rather than silently dropping. AUTHORED

beta

EDGE REAL – 0 100%

EDGE-rung coupling coefficient – `readouts.run(level='arm')`, the same-seed collapse REMOVED. AUTHORED

dGlobal_HLY_NAT ARMREAL, SIGNED63%	in the GTEx-healthy state $\hat{A}$ in matched normal (NAT) DERIVED Defined on: arms with a GTEx-HEALTHY leg (the thinnest state $\hat{A}$ many arms are multi-mapping-collapsed, MH-166)
dGlobal_HLY_TUM ARMREAL, SIGNED63%	in the GTEx-healthy state $\hat{A}$ in tumour DERIVED Defined on: arms with a GTEx-HEALTHY leg (the thinnest state $\hat{A}$ many arms are multi-mapping-collapsed, MH-166)
dGlobal_NAT_TUM ARMREAL, SIGNED68%	in matched normal (NAT) $\hat{A}$ in tumour DERIVED Defined on: edges with a matched NAT leg (n~104 paired participants)
dShare_M_own EDGE SIGNED $\hat{A}^{\prime}1\hat{A}\in\{1$65%	Mean over the 103 paired patients of this arm's change in M-WEIGHTED share of its gene's total regulator dose, tumour minus that patient's OWN NAT. AUTHORED
dShare_raw_own EDGE SIGNED $\hat{A}^{\prime}1\hat{A}\in\{1$65%	As dShare_M_own but on RAW abundance share, with the model's M weights removed. AUTHORED
gene KEYIDENTIFIER100%	KEY $\hat{A}$ HGNC gene symbol. AUTHORED
identified EDGEBOOLEAN100%	$ z > 2$ on the UNCALIBRATED posterior SD, where $z = \text{beta} / \text{beta_sd}$ (MH-83 records precision 0.86 / recall 0.89 against a held-out standard). True on 24.8% of edges. AUTHORED
identity EDGEREAL, SIGNED96%	Shapley/LMG share of the gene's R^2 credited to this edge, with bagged-NNLS weights. AUTHORED
z EDGEREAL $\hat{A}\%0$100%	$\text{beta} / \text{beta_sd}$ for the EDGE-level fit (<code>readouts.run(level='arm')</code>), the whole gene's arms as separate predictors), on the UNCALIBRATED posterior SD. AUTHORED

COUPLING_ $\hat{A}$ 13

core $\hat{A}$ the columns findings rest on

Partial-Spearman coupling of arm dose against target mRNA in one state, conditioned on C (composition + target CN + proliferation), with its p and z.

13 columns $\hat{A}$ coverage **3%~~100%~~** (median 62%) $\hat{A}$ **9** share the block description

5x signed $\hat{A}^{\prime}1\hat{A}\in\{1$ 5x p-value $\hat{A}$ 3x real, signed

coupling_fam FAMILY SIGNED $\hat{A}^{\prime}1\hat{A}\in\{1$100%	Family-level coupling for the edge's (gene, seed_family) cell. AUTHORED
coupling_hly EDGE SIGNED $\hat{A}^{\prime}1\hat{A}\in\{1$56%	in the GTEx-healthy state DERIVED Defined on: arms with a GTEx-HEALTHY leg (the thinnest state $\hat{A}$ many arms are multi-mapping-collapsed, MH-166)
coupling_hly_surrogate EDGE SIGNED $\hat{A}^{\prime}1\hat{A}\in\{1$3%	Healthy coupling via a same-seed SURROGATE arm, used where the arm itself has no GTEx signal. AUTHORED
coupling_nat EDGE SIGNED $\hat{A}^{\prime}1\hat{A}\in\{1$62%	in matched normal (NAT) DERIVED Defined on: edges with a matched NAT leg (n~104 paired participants)
coupling_p_hly EDGE P-VALUE56%	p-value $\hat{A}$ in the GTEx-healthy state DERIVED Defined on: arms with a GTEx-HEALTHY leg (the thinnest state $\hat{A}$ many arms are multi-mapping-collapsed, MH-166)

coupling_p_hly_resolved EDGE P-VALUE 59%	p-value $\hat{A}$ in the GTEx-healthy state DERIVED Defined on: arms with a same-seed SURROGATE (MH-166) $\hat{A}$ 3.6% of edges
coupling_p_hly_surrogate EDGE P-VALUE 3%	p-value $\hat{A}$ in the GTEx-healthy state DERIVED Defined on: arms with a same-seed SURROGATE (MH-166) $\hat{A}$ 3.6% of edges
coupling_p_nat EDGE P-VALUE 62%	p-value $\hat{A}$ in matched normal (NAT) DERIVED Defined on: edges with a matched NAT leg (n~104 paired participants)
coupling_p_tum EDGE P-VALUE 69%	p against the CALIBRATED site-free null (not a theoretical t-null). Median 0.375; <0.05 on 16.2%. AUTHORED
coupling_tum EDGE SIGNED $\hat{A}^{'1}\hat{A} 1$ 72%	in tumour DERIVED Defined on: edges present in the progression panel (arm expressed + state measured)
coupling_z_hly EDGE REAL, SIGNED 59%	z-score $\hat{A}$ in the GTEx-healthy state DERIVED Defined on: arms with a GTEx-HEALTHY leg (the thinnest state $\hat{A}$ many arms are multi-mapping-collapsed, MH-166)
coupling_z_nat EDGE REAL, SIGNED 62%	z-score $\hat{A}$ in matched normal (NAT) DERIVED Defined on: edges with a matched NAT leg (n~104 paired participants)
coupling_z_tum EDGE REAL, SIGNED 69%	The tumour coupling expressed in NULL SDs $\hat{A}$ the CALIBRATED effect size, and the scale to quote. Median $\hat{A}^{'0.32}$; z >2 on 15.5%, z >3 on 6.4%. AUTHORED

SEED_ $\hat{A}$ 1

core $\hat{A}$ the columns findings rest on

1 columns $\hat{A}$ coverage 100% $\hat{A}$ 100% (median 100%) 1x identifier

seed_family FAMILY IDENTIFIER 100%	KEY $\hat{A}$ the seed family itself, one row per family. AUTHORED
--	--

ARM_ $\hat{A}$ 15

core $\hat{A}$ the columns findings rest on

15 columns $\hat{A}$ coverage 20% $\hat{A}$ 100% (median 69%) 4x real $\hat{A}$ 4x real, signed $\hat{A}$ 3x fraction $\hat{A}$ 1 $\hat{A}$ 2x categorical $\hat{A}$ 1x percent $\hat{A}$ 1x boolean

arm_credit_share ARM-IN-FAMILY FRACTION $\hat{A}$ 1 20%	This arm's share of its (gene, seed_family) cell's credit. $\hat{A}$... Verified to sum to 1.0000 within every one of the 467 multi-arm cells, and it genuinely varies PER ARM inside the cell (467/467). AUTHORED Defined on: multi-arm cells only ($n_{arm_in_cell} > 1$) $\hat{A}$ 20.3% of edges
arm_dbeta FAMILY REAL $\hat{A}$ 20%	$\max \hat{A}^{'min} \hat{I}^2$ across the cell's member arms. AUTHORED
arm_detection ARM FRACTION $\hat{A}$ 1 95%	Fraction of patients in which this ARM is measured at all (readouts._detection). AUTHORED
arm_id_status ARM-IN-FAMILY CATEGORICAL 100%	How the arm's identity within its cell was resolved: resolved_by_dose 2,344 $\hat{A}$ singleton 2,260 $\hat{A}$ unvalidated_balanced 1,045. AUTHORED

arm_iqr ARMREAL 69%	Interquartile range of the arm's abundance – its DYNAMIC RANGE. AUTHORED
arm_lfc_HLY_NAT_raw ARMREAL, SIGNED63%	log-FC GTEx-healthy – TCGA-NAT, raw. AUTHORED
arm_lfc_HLY_TUM_QN ARMREAL, SIGNED64%	log-FC GTEx-healthy – TCGA-tumour, on quantile-normalised values. AUTHORED
arm_lfc_HLY_TUM_raw ARMREAL, SIGNED63%	As arm_lfc_HLY_TUM_QN but against the RAW GTEx median. AUTHORED
arm_lfc_NAT_TUM ARMREAL, SIGNED68%	log-FC of the arm's median abundance, matched NAT – tumour. AUTHORED
arm_med_rpm ARMREAL 69%	Median linear RPM of this ARM across the cohort. Arm-rung, so it REPEATS across the arm's genes. Fill 0.686 – the gap is arms with no expression record, not zeros. AUTHORED Defined on: edges present in the progression panel (arm expressed + state measured)
arm_pct_above_floor ARMPERCENT69%	Percent of samples in which the arm sits above the detection floor (median 99). Feeds arm_spiker together with arm_iqr. AUTHORED Defined on: edges present in the progression panel (arm expressed + state measured)
arm_role_in_family ARM-IN-FAMILYCATEGORICAL72%	– The ARM's DOSE role inside its seed family – NOT the family's role in the gene, which is what the name suggests. AUTHORED
arm_sep_z FAMILYREAL 20%	arm_dbeta in units of the cell's own pooled posterior SD – can these same-seed arms be told apart at all. AUTHORED
arm_share_of_family_dose ARM-IN-FAMILYFRACTION 69%	The arm's share of its seed family's TOTAL dose – denominator is the full family , not the design (median 0.69). The quantity family_role is cut from. AUTHORED Defined on: edges present in the progression panel (arm expressed + state measured)
arm_spiker ARMBOOLEAN69%	arm_pct_floor < 40 AND arm_iqr > 1.5 – the arm sits at/below the detection floor in most samples yet swings widely when present. AUTHORED

ADM_ 4

core – the columns findings rest on

4 columns – coverage 87%–87% (median 87%) 4× boolean	
adm_admissible EDGEBOOLEAN87%	The admissibility verdict: site + evidence + expression. AUTHORED
adm_expr_gtex EDGEBOOLEAN87%	ADMISSIBILITY – could this edge repress in breast AT ALL? has_site (seed match in the 3'UTR) AND expressed (arm present in TCGA tumour OR GTEx healthy breast). AUTHORED
adm_expr_tcga EDGEBOOLEAN87%	Is the arm expressed above floor in TCGA? AUTHORED
adm_has_site EDGEBOOLEAN87%	Does the arm have a seed match in this target's 3'UTR. AUTHORED

BETA_ 8

core the columns findings rest on

8 columns coverage 20%100% (median 100%)
4x real 0 4x fraction 01

beta_arm	The arm's OWN $\hat{I}^2$ from the within-cell fit (<code>arm_rung.py</code>), defined only in multi-arm cells (fill 0.203). AUTHORED
beta_arm	EDGE REAL 0 20%
beta_deconv	$\hat{I}^2$ refit on the DECONV design (core C + the 8 non-malignant Wu-major lineages, Cancer-Epithelial held out). AUTHORED
beta_deconv	EDGE REAL 0 100%
beta_frac	MAGNITUDE share (<code>beta_f</code> / sum beta). AUTHORED
beta_frac	EDGE FRACTION 01 100%
beta_frac_abs	$ E[\hat{I}^2_f] / E[\hat{I}^2] $ the share computed from the POINT ESTIMATES. AUTHORED
beta_frac_abs	EDGE FRACTION 01 100%
beta_frac_deconv	<code>beta_frac</code> recomputed on the deconv design the composition-adjusted share of the gene's budget. AUTHORED
beta_frac_deconv	EDGE FRACTION 01 100%
beta_frac_sd	Posterior SD of the per-draw fraction $\hat{I}^2_f / \hat{I}^2$. AUTHORED
beta_frac_sd	EDGE FRACTION 01 100%
beta_sd	Posterior summaries of the coupling coefficient from the dense Gibbs fit (<code>learned/readouts.py</code>). <code>_sd</code> is the posterior SD, <code>_frac</code> the share of the gene's total beta. AUTHORED
beta_sd	EDGE REAL 0 100%
beta_sd_deconv	Posterior SD on the deconv design. AUTHORED
beta_sd_deconv	EDGE REAL 0 100%

N_ 5

core the columns findings rest on

5 columns coverage 20%100% (median 100%)
4x count 1x constant

n_HLY_meas	How many of this gene's arms have a MEASURED healthy (GTEx) level as opposed to one manufactured by the multi-mapping collapse. AUTHORED
n_HLY_meas	EDGE COUNT 72%
n_arm_in_cell	How many arms sit in this edge's (gene, seed_family) CELL i.e. how many of that family's members appear in THIS gene's design. AUTHORED
n_arm_in_cell	FAMILY COUNT 20%
n_fam	How many families compete at this edge's GENE. Gene-rung it repeats across the gene's edges by construction. beta is mathematically inert where it is 1. AUTHORED
n_fam	GENE COUNT 100%
n_family_in_design	Members of this arm's seed family THAT ARE IN THIS GENE'S DESIGN 1 on 4,497 of 5,644 edges (79.7%) . AUTHORED
n_family_in_design	FAMILY COUNT 100%
n_samples	Samples the edge was fit on CONSTANT (1,040) across the card. AUTHORED
n_samples	GENE CONSTANT 100%

SHARE_ 4

core the columns findings rest on

The arm's share of the gene's total regulatory dose in one state (HLY / NAT / TUM). Within-gene, sums to 1 across the gene's arms.
4 columns coverage 46%72% (median 72%) 2 share the block description
4x fraction 01

share_HLY	The arm's share of the gene's healthy (GTEx) pressure budget, median 0.129. AUTHORED
EDGE	FRACTION 0172%
share_HLY_meas	share_HLY over ONLY the arms GTEx genuinely measures invisible arms ABSTAIN (NaN) instead of being zeroed (MH-210). Median 0.244 against the imputed 0.129. AUTHORED
EDGE	FRACTION 0146%
share_NAT	in matched normal (NAT) DERIVED
EDGE	FRACTION 0172% Defined on: edges with a matched NAT leg (n~104 paired participants)
share_TUM	in tumour DERIVED
EDGE	FRACTION 0172% Defined on: edges present in the progression panel (arm expressed + state measured)

CAL_ 7

core the columns findings rest on

CALIBRATED posterior width (learned/calibration.py): the reported SD corrected for the measured 1.181.34x narrowness, with _lo / _hi spanning the correction constant's own +/-1 SD.
7 columns coverage 100%100% (median 100%) 4 share the block description
4x real 0 3x boolean

cal_beta_sd	standard deviation DERIVED
EDGE	REAL 0100%
cal_beta_sd_hi	standard deviation DERIVED
EDGE	REAL 0100%
cal_beta_sd_lo	standard deviation DERIVED
EDGE	REAL 0100%
cal_identified	Is the edge identified (cal_z > 2) under the CALIBRATED posterior width the honest version of identified : 1,110 vs 1,375 edges, i.e. 19.9% [18.122.4] against 24.8%. AUTHORED
EDGE	BOOLEAN 100%
cal_identified_hi	The UPPER end of the calibrated identification band cal_z > 2 under the optimistic calibration (990 edges vs 1,110 central). AUTHORED
EDGE	BOOLEAN 100%
cal_identified_lo	The LOWER end of the calibrated identification band (1,234 edges vs 1,110 central). See cal_identified_hi the pair is the uncertainty, and reporting one alone hides it. AUTHORED
EDGE	BOOLEAN 100%
cal_z	z-score DERIVED
EDGE	REAL 0100%

ECHIM_ 3

core the columns findings rest on

3 columns coverage 8%26% (median 22%)
2x count 1x real 0

echim_manakov_w

Manakov chimeric-eCLIP support weight for this edge (median 3, max 1,712). AUTHORED

EDGE REAL 0 22%

echim_n_sources

How many chimeric-eCLIP sources support this edge (1 on 1,151 2 on 262 3 on 34). AUTHORED

EDGE COUNT 26%

echim_tarbase_w

TarBase chimeric support weight (median 2, max 28) a much shorter tail than the Manakov twin, and defined on only 8.4% of edges. AUTHORED

EDGE COUNT 8%

Defined on: edges with a chimeric (AGO-ligation) duplex as per-source, NEVER pooled; cross-tissue, not breast

SHIFT_ 1

core the columns findings rest on

1 columns coverage 72%72% (median 72%)
1x categorical

shift_class

Cross-state class for the edge. AUTHORED

EDGE CATEGORICAL 72%

RANK_ 4

core the columns findings rest on

The arm's WITHIN-GENE rank by dose in one state.

4 columns coverage 46%72% (median 72%) 4 share the block description
4x count

rank_HLY

in the GTEx-healthy state DERIVED

EDGE COUNT 72%

Defined on: arms with a GTEx-HEALTHY leg (the thinnest state many arms are multi-mapping-collapsed, MH-166)

rank_HLY_meas

in the GTEx-healthy state measured only (un-imputed) DERIVED

EDGE COUNT 46%

Defined on: edges whose gene has 1 GTEx-MEASURED regulator as NaN = GTEx cannot see it (abstention), NOT 0

rank_NAT

in matched normal (NAT) DERIVED

EDGE COUNT 72%

Defined on: edges with a matched NAT leg (n~104 paired participants)

rank_TUM

in tumour DERIVED

EDGE COUNT 72%

Defined on: edges present in the progression panel (arm expressed + state measured)

HEALTHY_ 3

supporting read these when the core raises a question

3 columns coverage 64%72% (median 64%)
1x categorical 1x real 0 1x boolean

healthy_leg ARM CATEGORICAL 72%	Provenance of this arm's GTEx healthy leg â€” measured , collapse_blind or true_absent . AUTHORED
healthy_potential ARM REAL 0 64%	The arm's IMPUTED healthy abundance â€” a repression-POTENTIAL proxy, collapse-repaired (0.27 â€” 17.9). AUTHORED
healthy_uninformative ARM BOOLEAN 64%	Is the healthy leg both collapse_blind AND high-potential â€” i.e. the one combination where an *acquired* call is genuinely unsafe? True on 143 of 3,490 edges (4%). AUTHORED

CTX_ 18
supporting â€” read these when the core raises a question

Gene-design context from the decoy/atlas layer (learned/analyses/gene_atlas.py via card_context): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.

18 columns 1 coverage 84%100% (median 99%) 10 share the block description
4x real 0 4x signed 11 3x real, signed 3x count 2x boolean 1x categorical 1x fraction 01

ctx_apriori_class GENE CATEGORICAL 100%	class label DERIVED
ctx_arm_abundant ARM BOOLEAN 95%	Is this ARM above the design's abundance threshold. AUTHORED
ctx_arm_dose ARM REAL 0 95%	The ARM's dose in this gene's design context. AUTHORED
ctx_ceiling GENE SIGNED 11 99%	The UPPER BOUND on what ANY estimator can achieve for this gene â€” the 5-fold out-of-fold R ² of an UNRESTRICTED OLS on the gene's real design (card_context.py:15). AUTHORED
ctx_collapse GENE REAL 0 100%	n_arm / n_fam for the gene â€” how much curation BUNDLES same-seed mates onto it (1â€”5). AUTHORED
ctx_d_collin GENE SIGNED 11 84%	Within-design COLLINEARITY asymmetry, real minus decoy. The second design-match axis MH-167 tested; also bounded and not moving the gap. AUTHORED Defined on: genes with >=2 seed families (a collinearity measure needs 2 columns)
ctx_d_fam GENE REAL, SIGNED 99%	Design-WIDTH asymmetry between the real design and its matched decoy (real minus fake families). AUTHORED
ctx_dose_max GENE REAL 0 100%	dose 1 maximum DERIVED
ctx_dose_med GENE REAL 0 100%	dose 1 median DERIVED
ctx_frac_abund GENE FRACTION 01 100%	fraction 1 unweighted abundance-sum reference DERIVED
ctx_gap_core GENE SIGNED 11 99%	gap 1 core confounder block DERIVED
ctx_gap_d_dose GENE REAL, SIGNED 99%	Dose asymmetry between the real and decoy designs (median 10.43). AUTHORED

ctx_gap_deconv	gap · composition-adjusted (deconvolution block)	DERIVED
GENE	SIGNED $\hat{\cdot}^1$	99%
ctx_gap_retention	gap · retention	DERIVED
GENE	REAL, SIGNED	95%
ctx_measurable	Is the gene's ctx_ceiling above CEILING_ROBUST (0.02, not 0 · MH-144)?	AUTHORED
GENE	BOOLEAN	99%
ctx_n_abund	count · unweighted abundance-sum reference	DERIVED
GENE	COUNT	100%
ctx_n_fam_abund	count · per seed family · unweighted abundance-sum reference	DERIVED
GENE	COUNT	100%
ctx_n_fam_multi	count · per seed family	DERIVED
GENE	COUNT	100%

IDENTITY_ · 5

supporting · read these when the core raises a question

5 columns · coverage 96%–98% (median 96%)
2× fraction 0–1 · 2× real, signed · 1× boolean

identity_abs	identity as a fraction of the gene's total identity , clipped to [0,1].	AUTHORED
EDGE	FRACTION 0–1	96%
identity_allocated	Family identity with phantom minor arms forced to 0 · the allocation actually used downstream.	AUTHORED
EDGE	REAL, SIGNED	96%
identity_coherence	1 / Î identity over the gene, clipped to (0,1] · how much the gene's identity shares CANCEL.	AUTHORED
EDGE	FRACTION 0–1	96%
identity_deconv	Shapley identity recomputed under the composition-adjusted design.	AUTHORED
EDGE	REAL, SIGNED	98%
identity_reliable	Is this edge's Shapley identity trustworthy · identity_coherence · 0.5 AND identity · 1?	AUTHORED
EDGE	BOOLEAN	96%

DOSE_ · 7

supporting · read these when the core raises a question

Dose (abundance) of the regulator and how it shifts across states, plus host-gene co-transcription fractions where the arm is intragenic.
7 columns · coverage 55%–72% (median 72%) · 6 share the block description
3× count · 2× real · 0 · 1× categorical · 1× constant

dose_class	class label	DERIVED
EDGE	CATEGORICAL	72%
dose_comp_retention	retention	DERIVED
ARM	REAL · 0	55%

dose_confounded ARM CONSTANT 72%	Is this edge's NAT's tumour dose a composition/proliferation artifact (either gated retention < 0.4)? AUTHORED
dose_prolif_retention ARM REAL 55%	retention DERIVED Defined on: edges with a matched NAT leg (n~104 paired participants)
dose_rank_HLY EDGE COUNT 72%	rank in the GTEx-healthy state DERIVED Defined on: arms with a GTEx-HEALTHY leg (the thinnest state many arms are multi-mapping-collapsed, MH-166)
dose_rank_NAT EDGE COUNT 72%	rank in matched normal (NAT) DERIVED Defined on: edges with a matched NAT leg (n~104 paired participants)
dose_rank_TUM EDGE COUNT 72%	rank in tumour DERIVED Defined on: edges present in the progression panel (arm expressed + state measured)

D_ 4
supporting read these when the core raises a question

Change in the arm's WITHIN-GENE dose rank between two states.

4 columns coverage 46%72% (median 72%) 3 share the block description
4x real, signed

d_rank_HLY_NAT EDGE REAL, SIGNED 72%	rank in the GTEx-healthy state in matched normal (NAT) DERIVED Defined on: arms with a GTEx-HEALTHY leg (the thinnest state many arms are multi-mapping-collapsed, MH-166)
d_rank_HLY_TUM EDGE REAL, SIGNED 72%	rank in the GTEx-healthy state in tumour DERIVED Defined on: arms with a GTEx-HEALTHY leg (the thinnest state many arms are multi-mapping-collapsed, MH-166)
d_rank_HLY_TUM_meas EDGE REAL, SIGNED 46%	d_rank_HLY_TUM recomputed over arms with a MEASURED healthy level only the collapse-safe version. Prefer it over the bare column. AUTHORED Defined on: edges whose gene has a GTEx-MEASURED regulator's NaN = GTEx cannot see it (abstention), NOT 0
d_rank_NAT_TUM EDGE REAL, SIGNED 72%	rank in matched normal (NAT) in tumour DERIVED Defined on: edges with a matched NAT leg (n~104 paired participants)

GENE_ 6
supporting read these when the core raises a question

Gene-level properties carried onto this row (its regulator count, repression class, net-repression flag). On the edge card these REPEAT within gene by construction.

6 columns coverage 72%100% (median 99%) 3 share the block description
2x categorical 2x boolean 1x real 1x real, signed

gene_cancer_role GENE CATEGORICAL 100%	NOT the arm's role the GENE's cancer role (oncogene / tsg / oncogene/tsg / unknown), from gene_roles.load_gene_roles. AUTHORED
gene_dominated GENE BOOLEAN 97%	Does one family carry more than 60% of this gene's tumour pressure budget (2,641 of 5,495)? AUTHORED

gene_iqr GENE REAL $\in [0, 1]$ 72%	interquartile range DERIVED Defined on: genes with a NAT-TUM gene-level leg
gene_lfc_NAT_TUM GENE REAL, SIGNED 72%	in matched normal (NAT) Δ in tumour DERIVED Defined on: genes with a NAT-TUM gene-level leg
gene_net_repr GENE BOOLEAN 100%	Is this gene NET-REPRESSED in tumour by the Δ 6b-RETIRED heuristic lane (1,376 of 5,634 edges)? AUTHORED
gene_repr_class GENE CATEGORICAL 100%	class label DERIVED

AGO_ Δ 2

supporting Δ read these when the core raises a question

2 columns Δ coverage 95%Δ100% (median 97%) 1x fraction $\in [0, 1]$ Δ 1x boolean	
ago_loading ARM FRACTION $\in [0, 1]$ 95%	AGO/RISC loading discount. AUTHORED
ago_loading_measured ARM BOOLEAN 100%	Was <code>ago_loading</code> actually MEASURED for this edge (vs defaulted to 1.0). True for 3,820/5,648 edges (67.6%). AUTHORED Defined on: arms with a Manakov AGO-dominance measurement Δ NOT loading(), which fabricates 1.0

PIP_ Δ 2

supporting Δ read these when the core raises a question

2 columns Δ coverage 100%Δ100% (median 100%) 1x constant Δ 1x fraction $\in [0, 1]$	
pip_dense EDGE CONSTANT 100%	Δ CONSTANT by construction Δ the coupling readout is called with $\in [0, 1]$, so every edge has the same value. AUTHORED
pip_discovery EDGE FRACTION $\in [0, 1]$ 100%	Posterior inclusion probability. <code>pip_dense</code> comes from the $\pi=1$ coupling readout; <code>pip_discovery</code> from the evidence- π readout. AUTHORED

RETENTION_ Δ 3

reference Δ context, provenance and rarely-quoted detail

3 columns Δ coverage 72%Δ100% (median 100%) 1x real $\in [0, 1]$ Δ 1x boolean Δ 1x real, signed	
retention_beta EDGE REAL $\in [0, 1]$ 100%	Fraction of an effect surviving covariate adjustment (adjusted / raw). AUTHORED
retention_reliable EDGE BOOLEAN 100%	Is this edge's <code>retention</code> ratio trustworthy Δ i.e. does it have a real denominator? AUTHORED
retention_rho EDGE REAL, SIGNED 72%	$\bar{I}_{\text{deconv}} / \bar{I}_{\text{core}}$ Δ AUTHORED

OOF_ 3

reference “ context, provenance and rarely-quoted detail

Out-of-fold predictive correlation, arm-level vs family-level, and their difference (`oof_drho` = arm minus family). Negative = the arm-resolved fit predicts repression better.

3 columns coverage 20%–20% (median 20%) 2 share the block description

3× signed 11

oof_drho FAMILY SIGNED 11 20%	Arm-level OOF rho MINUS family-level. AUTHORED
oof_rho_arm FAMILY SIGNED 11 20%	Spearman per arm DERIVED Defined on: multi-arm cells only ($n_{arm_in_cell} > 1$) 20.3% of edges
oof_rho_fam FAMILY SIGNED 11 20%	Spearman per seed family DERIVED Defined on: multi-arm cells only ($n_{arm_in_cell} > 1$) 20.3% of edges

GRANK_ 3

reference “ context, provenance and rarely-quoted detail

The arm's GLOBAL (cohort-wide) dose rank in one state. This is what the dominant-regulator and handoff columns are computed from.

3 columns coverage 63%–69% (median 68%) 3 share the block description

3× count

grank_HLY ARM COUNT 63%	in the GTEx-healthy state DERIVED Defined on: arms with a GTEx-HEALTHY leg (the thinnest state many arms are multi-mapping-collapsed, MH-166)
grank_NAT ARM COUNT 68%	in matched normal (NAT) DERIVED Defined on: edges with a matched NAT leg ($n \sim 104$ paired participants)
grank_TUM ARM COUNT 69%	in tumour DERIVED Defined on: edges present in the progression panel (arm expressed + state measured)

OWN_ 1

reference “ context, provenance and rarely-quoted detail

1 columns coverage 88%–88% (median 88%)

1× real 0

own_specific_frac

ARM

REAL 0

88%

Fraction of the arm's variance that is target-specific.

AUTHORED

COMPOSITION_ 1

reference " context, provenance and rarely-quoted detail

1 columns coverage 100%100% (median 100%)
1x categorical

composition_class

Composition-adjustment verdict for this row (e.g. composition_explained when the effect does not survive the deconvolution block). AUTHORED

EDGE

CATEGORICAL

100%

W_ 1

reference " context, provenance and rarely-quoted detail

1 columns coverage 100%100% (median 100%)
1x real 0

w_max

Maximum curated EVIDENCE weight over the gene's regulators. Gene-rung repeat. AUTHORED

GENE

REAL 0

100%

M_ 1

reference " context, provenance and rarely-quoted detail

1 columns coverage 100%100% (median 100%)
1x real 0

m_nnl

Bagged-NNLS weight. Exactly 0 in 31.7% of families " this is what lets identity say a regulator contributes nothing, which beta structurally cannot. AUTHORED

EDGE

REAL 0

100%

PRIOR_ 1

reference " context, provenance and rarely-quoted detail

1 columns coverage 100%100% (median 100%)
1x constant

prior_pi

The Gibbs sampler's inclusion prior " CONSTANT 0.05 across the card. AUTHORED

EDGE

CONSTANT

100%

EDGE_ 1

reference " context, provenance and rarely-quoted detail

1 columns coverage 73%73% (median 73%)
1x signed 11

edge_rho_adj

An edge-rung quantity carried onto this row. AUTHORED

EDGE

SIGNED 11

73%

| Defined on: the WITHIN-PATIENT paired subset (n=103 matched tumour/NAT pairs, MH-158)

TERM_ 3

reference " context, provenance and rarely-quoted detail

3 columns coverage 68%68% (median 68%) 2x signed 1 1 1x real, signed		
term_ABUND	Decomposition of the acquisition score into ABUNDANCE, WIRING and INTERACTION terms. AUTHORED	
EDGE	SIGNED 1 1	68%
Defined on: edges present in the progression panel (arm expressed + state measured)		
term_INTERACT	The INTERACTION term " the part of the NAT'tumour change attributable to abundance and wiring moving TOGETHER. AUTHORED	
EDGE	SIGNED 1 1	68%
term_WIRING	The WIRING term of the dose-vs-wiring decomposition: how much of the NAT'tumour change comes from the arm's M-weight moving rather than its abundance. AUTHORED	
EDGE	REAL, SIGNED	68%

MEAN_ 2

reference " context, provenance and rarely-quoted detail

2 columns coverage 90%90% (median 90%) 2x real, signed		
mean_dGlobalRank	Mean change in the arm's GLOBAL abundance percentile across the paired patients. AUTHORED	
ARM	REAL, SIGNED	90%
mean_own_shift	A per-row mean of a lower-rung quantity; check agg_of for which rung it summarises. AUTHORED	
ARM	REAL, SIGNED	90%
Defined on: the WITHIN-PATIENT paired subset (n=103 matched tumour/NAT pairs, MH-158)		

SURROGATE_ 2

reference " context, provenance and rarely-quoted detail

2 columns coverage 3%3% (median 3%) 1x fraction 1 1x categorical		
surrogate_corr	How well the surrogate tracks the arm where both are observed (0.54 " 0.89). AUTHORED	
ARM	FRACTION 1	3%
surrogate_instrument	Which SURROGATE stands in for a collapsed GTEx healthy leg (7 levels; defined on 3% of edges). AUTHORED	
ARM	CATEGORICAL	3%

ESUB_ 8

reference " context, provenance and rarely-quoted detail

PAM50 subtype-stratified coupling at the EDGE rung " one fit per subtype plus the pooled fit and an adjusted q. 8 columns coverage 98%100% (median 99%) 5 share the block description 6x signed 1 1 1x categorical 1x p-value		
esub_best_sub	the best DERIVED	
EDGE	CATEGORICAL	100%
Defined on: edges with a per-PAM50 heterogeneity fit " " DISTRIBUTIONAL (MH-165), not a per-edge subtype label		

esub_q_adj EDGE P-VALUE 98%	BH-adjusted q for the best-subtype contrast AUTHORED
esub_rho_Basal EDGE SIGNED $\hat{\rho}^1$ 99%	Spearman $\hat{\rho}$ DERIVED Defined on: edges with a per-PAM50 heterogeneity fit $\hat{\rho}$ DISTRIBUTIONAL (MH-165), not a per-edge subtype label
esub_rho_Her2 EDGE SIGNED $\hat{\rho}^1$ 98%	Spearman $\hat{\rho}$ DERIVED Defined on: edges with a per-PAM50 heterogeneity fit $\hat{\rho}$ DISTRIBUTIONAL (MH-165), not a per-edge subtype label
esub_rho_LumA EDGE SIGNED $\hat{\rho}^1$ 99%	Spearman $\hat{\rho}$ DERIVED Defined on: edges with a per-PAM50 heterogeneity fit $\hat{\rho}$ DISTRIBUTIONAL (MH-165), not a per-edge subtype label
esub_rho_LumB EDGE SIGNED $\hat{\rho}^1$ 99%	Spearman $\hat{\rho}$ DERIVED Defined on: edges with a per-PAM50 heterogeneity fit $\hat{\rho}$ DISTRIBUTIONAL (MH-165), not a per-edge subtype label
esub_rho_best EDGE SIGNED $\hat{\rho}^1$ 100%	Coupling in the arm's BEST PAM50 subtype. AUTHORED
esub_rho_pool EDGE SIGNED $\hat{\rho}^1$ 100%	Coupling pooled across subtypes $\hat{\rho}$ the unselected reference for esub_rho_best (median $\hat{\rho}$ 0.0135, indistinguishable from a random-subtype draw at $\hat{\rho}$ 0.0123). AUTHORED

REALIZATION_ 2

reference $\hat{\rho}$ context, provenance and rarely-quoted detail

2 columns $\hat{\rho}$ coverage 58%69% (median 63%) 1x fraction $\hat{\rho}$ 1x real, signed	
realization_identity EDGE FRACTION $\hat{\rho}$ 58%	Within-patient realization quantities $\hat{\rho}$ the score, its identity allocation, and who owns it. AUTHORED Defined on: genes with a family-level realization fit
realization_score EDGE REAL, SIGNED 69%	$\hat{\rho}$ coupling_z_tum $\hat{\rho}$ the calibrated tumour repression strength as a positive score, so bigger = more repressed. AUTHORED

WIRING_ 1

reference $\hat{\rho}$ context, provenance and rarely-quoted detail

1 columns $\hat{\rho}$ coverage 68%68% (median 68%) 1x fraction $\hat{\rho}$	
wiring_frac EDGE FRACTION $\hat{\rho}$ 68%	Fraction of the acquisition score attributable to WIRING rather than abundance. AUTHORED Defined on: edges present in the progression panel (arm expressed + state measured)

KD_ 2

reference " context, provenance and rarely-quoted detail

2 columns coverage 85%85% (median 85%)
1x percent 1x real, signed

kd_affinity_pct

EDGE PERCENT 85%

Is this gene among the arm's strongest predicted targets " a WITHIN-ARM percentile, **higher = stronger target.** AUTHORED

kd_repression

EDGE REAL, SIGNED 85%

scanMiR RBNS binding affinity. AUTHORED

CELL_ 1

reference " context, provenance and rarely-quoted detail

1 columns coverage 20%20% (median 20%)
1x boolean

cell_arms_resolvable

FAMILY BOOLEAN 20%

Are the cell's arms separable AND does splitting them help out-of-fold. True for 31.7% of edges in multi-arm cells. AUTHORED

SD_ 1

reference " context, provenance and rarely-quoted detail

1 columns coverage 20%20% (median 20%)
1x real 0

sd_arm

EDGE REAL 0 20%

Posterior SD of the arm-level $\hat{\tau}^2$. AUTHORED

ACQUISITION_ 1

reference " context, provenance and rarely-quoted detail

1 columns coverage 59%59% (median 59%)
1x real, signed

acquisition_score

EDGE REAL, SIGNED 59%

Composite acquisition score across states. AUTHORED
Defined on: arms with a GTEx-HEALTHY leg (the thinnest state " many arms are multi-mapping-collapsed, MH-166)

Z_ 1

reference " context, provenance and rarely-quoted detail

1 columns coverage 20%20% (median 20%)
1x real 0

z_arm

EDGE REAL 0 20%

β_{arm} / sd_{arm} from the WITHIN-CELL fit (`arm_rung.py`) " the arm's z inside its (gene, seed_family) cell only, defined on the 20.3% of edges in multi-arm cells. AUTHORED

HEUR_ 1

reference " context, provenance and rarely-quoted detail

1 columns coverage 100%100% (median 100%)
1x count

heur_gene_nreg

Regulator count for this gene from the 6b-RETIRED heuristic lane.

AUTHORED

GENECOUNT100%

FAMILY_ 1

reference " context, provenance and rarely-quoted detail

1 columns coverage 69%69% (median 69%)
1x signed 11

family_rho_adj

Family-level properties carried onto this row (size, the arm's dose share of it, and its role).

AUTHORED

FAMILYSIGNED 1169%

CPTAC_ 20

reference " context, provenance and rarely-quoted detail

CPTAC protein channel.
20 columns coverage 45%79% (median 65%) 16 share the block description
12x signed 11 4x real, signed 2x constant 2x categorical

cptac_prosp_comp_class_prot

Composition class of this edge's PROTEIN coupling (cell_intrinsic / partial / composition_explained) in the prospective cohort.

AUTHORED

EDGECATEGORYAL59%

cptac_prosp_n

Patients in the prospective CPTAC cohort " **CONSTANT 101.**

AUTHORED

EDGECONSTANT79%

cptac_prosp_ret_prot

CPTAC prospective cohort (n% ^101) against protein

DERIVED

Defined on: genes/edges present in the CPTAC cohort

EDGEREAL, SIGNED59%

cptac_prosp_ret_rna

CPTAC prospective cohort (n% ^101) against mRNA

DERIVED

Defined on: genes/edges present in the CPTAC cohort

EDGEREAL, SIGNED61%

cptac_prosp_rho_disc

CPTAC prospective cohort (n% ^101) Spearman I against the discovery layer

DERIVED

Defined on: genes/edges present in the CPTAC cohort

EdgesIGNED 1178%

cptac_prosp_rho_disc_raw

CPTAC prospective cohort (n% ^101) Spearman I against the discovery layer UNADJUSTED " no confounder block

DERIVED

Defined on: genes/edges present in the CPTAC cohort

EdgesIGNED 1178%

cptac_prosp_rho_prot

CPTAC prospective cohort (n% ^101) Spearman I against protein

DERIVED

Defined on: genes/edges present in the CPTAC cohort

EdgesIGNED 1178%

cptac_prosp_rho_prot_raw

UNADJUSTED protein coupling, prospective.

AUTHORED

EdgesIGNED 1179%

cptac_prosp_rho_rna

CPTAC prospective cohort (n% ^101) Spearman I against mRNA

DERIVED

Defined on: genes/edges present in the CPTAC cohort

EdgesIGNED 1179%

cptac_prosp_rho_rna_raw EDGE SIGNED $\hat{r}^2 \in [1, 1]$ 79%	CPTAC prospective cohort (n=101) $\hat{r}^2$ Spearman $\hat{r}^2$ against mRNA $\hat{r}^2$ UNADJUSTED no confounder block DERIVED <i>Defined on: genes/edges present in the CPTAC cohort</i>
cptac_t105_comp_class_prot EDGE CATEGORICAL 45%	CPTAC TCGA-overlap cohort (n=105) $\hat{r}^2$ class label $\hat{r}^2$ against protein DERIVED <i>Defined on: genes/edges present in the CPTAC cohort</i>
cptac_t105_n EDGE CONSTANT 70%	Patients in the TCGA-overlap CPTAC cohort are CONSTANT 105 . See the prospective twin. AUTHORED
cptac_t105_ret_prot EDGE REAL, SIGNED 45%	CPTAC TCGA-overlap cohort (n=105) $\hat{r}^2$ against protein DERIVED <i>Defined on: genes/edges present in the CPTAC cohort</i>
cptac_t105_ret_rna EDGE REAL, SIGNED 49%	CPTAC TCGA-overlap cohort (n=105) $\hat{r}^2$ against mRNA DERIVED <i>Defined on: genes/edges present in the CPTAC cohort</i>
cptac_t105_rho_disc EDGE SIGNED $\hat{r}^2 \in [1, 1]$ 64%	CPTAC TCGA-overlap cohort (n=105) $\hat{r}^2$ Spearman $\hat{r}^2$ against the discovery layer DERIVED <i>Defined on: genes/edges present in the CPTAC cohort</i>
cptac_t105_rho_disc_raw EDGE SIGNED $\hat{r}^2 \in [1, 1]$ 65%	CPTAC TCGA-overlap cohort (n=105) $\hat{r}^2$ Spearman $\hat{r}^2$ against the discovery layer $\hat{r}^2$ UNADJUSTED no confounder block DERIVED <i>Defined on: genes/edges present in the CPTAC cohort</i>
cptac_t105_rho_prot EDGE SIGNED $\hat{r}^2 \in [1, 1]$ 64%	CPTAC TCGA-overlap cohort (n=105) $\hat{r}^2$ Spearman $\hat{r}^2$ against protein DERIVED <i>Defined on: genes/edges present in the CPTAC cohort</i>
cptac_t105_rho_prot_raw EDGE SIGNED $\hat{r}^2 \in [1, 1]$ 65%	CPTAC TCGA-overlap cohort (n=105) $\hat{r}^2$ Spearman $\hat{r}^2$ against protein $\hat{r}^2$ UNADJUSTED no confounder block DERIVED <i>Defined on: genes/edges present in the CPTAC cohort</i>
cptac_t105_rho_rna EDGE SIGNED $\hat{r}^2 \in [1, 1]$ 64%	CPTAC TCGA-overlap cohort (n=105) $\hat{r}^2$ Spearman $\hat{r}^2$ against mRNA DERIVED <i>Defined on: genes/edges present in the CPTAC cohort</i>
cptac_t105_rho_rna_raw EDGE SIGNED $\hat{r}^2 \in [1, 1]$ 65%	CPTAC TCGA-overlap cohort (n=105) $\hat{r}^2$ Spearman $\hat{r}^2$ against mRNA $\hat{r}^2$ UNADJUSTED no confounder block DERIVED <i>Defined on: genes/edges present in the CPTAC cohort</i>

HLY_ $\hat{r}^2$ 1

reference are context, provenance and rarely-quoted detail

1 columns $\hat{r}^2$ coverage 61%61% (median 61%) 1x boolean

hly_leg_concordant EDGE BOOLEAN 61%	Do the two HLY+TUM legs (_QN and _raw) at least AGREE ON SIGN. Concordant 76.1%, DISCORDANT 23.9% of the 3,433 edges carrying both. AUTHORED
--	--

IS_ $\hat{r}^2$ 1

reference are context, provenance and rarely-quoted detail

1 columns $\hat{r}^2$ coverage 58%58% (median 58%) 1x boolean

is_realization_owner EDGE BOOLEAN 58%	A boolean flag; read the suffix. AUTHORED <i>Defined on: genes with a family-level realization fit</i>
--	--

realization/gene_card.tsv

gene

One row per the gene's total incoming regulation. One row per gene, so nothing here is checkable by invariance – `agg_of` carries the meaning instead.

135 columns 1,549 rows 29 blocks median coverage 83%

Contents – 29 blocks, most important first

1. n_ 7 – CORE

2. (bare) 2 – CORE

3. comp_ 16 – CORE

4. ctx_ 16 – CORE

5. top_ 8 – SUPPORTING

6. realized_ 5 – SUPPORTING

7. w_ 1 – SUPPORTING

8. lit_ 7 – SUPPORTING

9. cptac_ 24 – REFERENCE

10. frac_ 3 – REFERENCE

11. dominant_ 4 – REFERENCE

12. static_ 1 – REFERENCE

13. total_ 1 – REFERENCE

14. owner_ 1 – REFERENCE
15. median_ 1 – REFERENCE

16. armres_ 7 – REFERENCE

17. identity_ 1 – REFERENCE

18. heur_ 5 – REFERENCE

19. greal_ 10 – REFERENCE

20. acquired_ 2 – REFERENCE

21. pressure_ 1 – REFERENCE

22. realization_ 1 – REFERENCE

23. tcga_ 4 – REFERENCE

24. disc_ 2 – REFERENCE

25. max_ 1 – REFERENCE

26. regulatory_ 1 – REFERENCE

27. mean_ 1 – REFERENCE

28. concentration_ 1 – REFERENCE

29. fam_ 1 – REFERENCE

N_ – 7

core – the columns findings rest on

7 columns – coverage 91%–100% (median 100%)
7x count

n_arms

GENE COUNT AGG ARM 100%

Arms in the LEARNED MODEL's design – `X.shape[1]` from `assemble_gene`, carried via `gene_atlas`. **This is the width the Gibbs actually fit** (median 2, max 90). AUTHORED

n_cell_intrinsic

GENE COUNT 100%

FAMILIES whose coupling SURVIVES the composition block (retention ≥ 0.7). AUTHORED

n_composition_explained

GENE COUNT 100%

FAMILIES whose coupling does NOT survive the composition block (retention < 0.4). AUTHORED

n_fam

GENE COUNT AGG FAMILY
100%

How many seed families regulate this gene. AUTHORED

n_hallmark_sets

GENE COUNT 91%

How many MSigDB Hallmark programs contain this gene (1–10, median 2). Membership only – it says nothing about regulation. AUTHORED

n_identified

GENE COUNT 100%

Families reaching $|z| > 2$ on the UNCALIBRATED SD. **Zero for 691 genes (44.6%)** – meaning the model cannot distinguish ANY of that gene's families from zero. AUTHORED

n_pip_disc_gt50 GENE COUNT 100%	Count of the gene's families with <code>pip_discovery > 0.5</code> under the evidence-<code>Ĭ</code> readout. AUTHORED
(BARE) 2 core the columns findings rest on	
2 columns coverage 100%100% (median 100%) 1x fraction 01 1x identifier	
concentration GENE FRACTION 01 100%	<code>beta.max() / beta.sum()</code> the TOP FAMILY's share of the gene's total $\hat{\pi}^2$; median 0.954, i.e. most genes are dominated by one family. AUTHORED
gene KEY IDENTIFIER 100%	KEY HGNC gene symbol. AUTHORED
COMP_ 16 core the columns findings rest on	
Cell-composition drivers of this gene's coupling (<code>compartment_*</code>): which deconvolved fraction explains the correlation, its retention under adjustment, and its share. 16 columns coverage 59%90% (median 73%) 7 share the block description 6x categorical 6x signed 11 3x real, signed 1x real 0	
comp_cptac_prot_driver GENE CATEGORICAL 59%	CPTAC against protein DERIVED Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)
comp_cptac_prot_driver_ret GENE REAL, SIGNED 59%	1 comp_cptac_prot_driver_share, exactly. Median +0.398 raw, +0.469 gated to <code> Ĭ_raw %</code> 0.10 quote the gated value. See the mRNA twin. AUTHORED Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)
comp_cptac_prot_driver_share GENE REAL, SIGNED 59%	Protein-layer twin of <code>comp_tcga_mrna_driver_share</code>. Same exact complementarity with <code>_driver_ret</code> (8.9e-16 on 918 rows). AUTHORED
comp_cptac_prot_rho_raw GENE SIGNED 11 73%	Unadjusted aggregate PROTEIN coupling denominator of <code>comp_cptac_prot_driver_ret</code>. AUTHORED
comp_cptac_prot_top_budget GENE CATEGORICAL 73%	CPTAC against protein the top-ranked pressure budget DERIVED Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)
comp_cptac_prot_top_budget_r GENE SIGNED 11 73%	CPTAC against protein the top-ranked pressure budget correlation DERIVED Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)
comp_cptac_prot_top_target GENE CATEGORICAL 73%	CPTAC against protein the top-ranked DERIVED Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)
comp_cptac_prot_top_target_r GENE SIGNED 11 73%	CPTAC against protein the top-ranked correlation DERIVED Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)
comp_tcga_mrna_driver GENE CATEGORICAL 66%	WHICH compartment drives the gene's mRNA coupling the one whose removal moves it most (<code>CAFs</code> 279 <code>purity</code> 174 <code>prolif</code> 148 <code>Endothelial</code> 122 <code>Myeloid</code> 109). AUTHORED

comp_tcga_mrna_driver_ret	1 $\hat{A}^*$ comp_tcga_mrna_driver_share, exactly (see there). The RETENTION orientation: what fraction of the coupling survives removing the driver compartment. AUTHORED
comp_tcga_mrna_driver_share	Share of the gene's mRNA coupling attributable to the single compartment that most changes it $\hat{A}^*$ axiom 8's composition GATE , the axis that separates the harm tails at p=0.00999 where the ensemble axis gives p=0.10. AUTHORED
comp_tcga_mrna_rho_raw	The gene's UNADJUSTED aggregate mRNA coupling $\hat{A}^*$ the DENOMINATOR of comp_tcga_mrna_driver_ret. AUTHORED
comp_tcga_mrna_top_budget	The compartment the gene's miRNA BUDGET loads on most $\hat{A}^*$ a property of the REGULATORS. AUTHORED
comp_tcga_mrna_top_budget_r	TCGA $\hat{A}^*$ the top-ranked $\hat{A}^*$ pressure budget $\hat{A}^*$ correlation DERIVED Defined on: genes/families with a TCGA-mRNA compartment driver $\hat{A}^*$ $\hat{A}^*$ MH-201: these axes predict the SIGN of $\hat{I}^2$'s gain
comp_tcga_mrna_top_target	The compartment the TARGET GENE's expression loads on most. AUTHORED
comp_tcga_mrna_top_target_r	TCGA $\hat{A}^*$ the top-ranked $\hat{A}^*$ correlation DERIVED Defined on: genes/families with a TCGA-mRNA compartment driver $\hat{A}^*$ $\hat{A}^*$ MH-201: these axes predict the SIGN of $\hat{I}^2$'s gain

CTX_ $\hat{A}^*$ 16
core $\hat{A}^*$ the columns findings rest on

Gene-design context from the decoy/atlas layer (learned/analyses/gene_atlas.py via card_context): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.

16 columns $\hat{A}^*$ coverage 45% $\hat{A}^*$ 100% (median 90%) $\hat{A}^*$ 11 share the block description
4x signed $\hat{A}^*$ 1 $\hat{A}^*$ 1 3x real $\hat{A}^*$ 0 $\hat{A}^*$ 3x real, signed $\hat{A}^*$ 3x count $\hat{A}^*$ 1x categorical $\hat{A}^*$ 1x fraction 0 $\hat{A}^*$ 1 $\hat{A}^*$ 1x boolean

ctx_apriori_class	A-priori competence class from the gene atlas (A_COMPETENT / B_PARTIAL / C_WEAK / D_UNDEFINED). AUTHORED
ctx_ceiling	The UPPER BOUND on what ANY estimator can achieve for this gene $\hat{A}^*$ the 5-fold out-of-fold $\hat{R}^2$ of an UNRESTRICTED OLS on the gene's real design (card_context.py:15). AUTHORED
ctx_collapse	n_arm / n_fam for the gene $\hat{A}^*$ how much curation BUNDLES same-seed mates onto it (1 $\hat{A}^*$ 5). AUTHORED
ctx_d_collin	change DERIVED Defined on: genes with ≥ 2 seed families (a collinearity measure needs 2 columns)
ctx_d_fam	change $\hat{A}^*$ per seed family DERIVED
ctx_dose_max	dose $\hat{A}^*$ maximum DERIVED
ctx_dose_med	dose $\hat{A}^*$ median DERIVED
ctx_frac_abund	fraction $\hat{A}^*$ unweighted abundance-sum reference DERIVED

ctx_gap_core GENE SIGNED $\hat{A}^1 \in \{1\}$ 87%	Real-minus-decoy coupling gap for this gene. Pooled it is ~ 0.012 , but the per-gene IQR STRADDLES ZERO $\hat{A}$ this is a set-level distributional shift, not a per-gene effect. Do not rank genes by it. AUTHORED
ctx_gap_d_dose GENE REAL, SIGNED 87%	gap $\hat{A}$ change $\hat{A}$ dose DERIVED
ctx_gap_deconv GENE SIGNED $\hat{A}^1 \in \{1\}$ 87%	gap $\hat{A}$ composition-adjusted (deconvolution block) DERIVED
ctx_gap_retention GENE REAL, SIGNED 83%	gap $\hat{A}$ retention DERIVED
ctx_measurable GENE BOOLEAN 98%	Is the gene's ctx_ceiling above CEILING_ROBUST (0.02 , not 0 $\hat{A}$ MH-144)? AUTHORED
ctx_n_abund GENE COUNT 90%	count $\hat{A}$ unweighted abundance-sum reference DERIVED
ctx_n_fam_abund GENE COUNT 90%	count $\hat{A}$ per seed family $\hat{A}$ unweighted abundance-sum reference DERIVED
ctx_n_fam_multi GENE COUNT 90%	count $\hat{A}$ per seed family DERIVED

TOP_ $\hat{A}$ 8
supporting $\hat{A}$ read these when the core raises a question

The gene's leading regulator and its value, by magnitude or by identity.

8 columns $\hat{A}$ coverage **78 $\hat{A}$ 100%** (median 89%) $\hat{A}$ **3** share the block description
3x identifier $\hat{A}$ 2x real $\hat{A} \in \{0\}$ $\hat{A}$ 1x fraction $\hat{A} \in \{1\}$ $\hat{A}$ 1x signed $\hat{A}^1 \in \{1\}$ $\hat{A}$ 1x count

top_beta GENE REAL $\hat{A} \in \{0\}$ AGG FAMILY 100%	$\hat{I}^2$ DERIVED
top_beta_frac GENE FRACTION $\hat{A} \in \{1\}$ AGG FAMILY 100%	$\hat{I}^2$ $\hat{A}$ fraction DERIVED
top_family_identity GENE IDENTIFIER AGG FAMILY 87%	The family Shapley/LMG credits with the gene's regulation. NaN for 197 genes (12.7%) where NNLS zeroed everything $\hat{A}$ an honest 'undefined', which the magnitude answer cannot express. AUTHORED
top_family_magnitude GENE IDENTIFIER AGG FAMILY 100%	The family with the largest beta. AUTHORED
top_gaining_arms GENE IDENTIFIER 91%	arms DERIVED
top_identity GENE REAL $\hat{A} \in \{0\}$ AGG FAMILY 87%	The largest identity share among the gene's families. AUTHORED

top_identity_gated	top_identity recomputed over identity_reliable edges only â€” max 1.0000, 0 genes above 1 (vs +740.007 ungated). Defined on 1,203 genes. AUTHORED
GENE SIGNED $\hat{\rho}^1 \in \{1\}$ 78%	
top_identity_n_reliable	How many of the gene's families survive the identity gate â€” the DENOMINATOR behind top_identity_gated . Median 2. AUTHORED
GENE COUNT 78%	

REALIZED_ $\hat{\rho}$ 5
supporting â€” read these when the core raises a question

Realized (as opposed to potential) coupling for the gene â€” the raw and adjusted rho, its retention, and how many regulators it was computed over.

5 columns $\hat{\rho}$ coverage **63%â€”82%** (median 81%) $\hat{\rho}$ **3** share the block description
2x signed $\hat{\rho}^1 \in \{1\}$ $\hat{\rho}$ 1x boolean $\hat{\rho}$ 1x count $\hat{\rho}$ 1x real, signed

realized_composition_explained	Is the gene's REALIZED coupling composition-explained (retention < 0.4)? AUTHORED
GENE BOOLEAN 82%	
realized_n_reg	count DERIVED
GENE COUNT 82%	
realized_retention	Realized-coupling retention, $\hat{\rho}_{\text{adjusted}} / \hat{\rho}_{\text{raw}}$. AUTHORED
GENE REAL, SIGNED 63%	
realized_rho_adj	Spearman $\hat{\rho}$ $\hat{\rho}$ composition-adjusted DERIVED
GENE SIGNED $\hat{\rho}^1 \in \{1\}$ 81%	
realized_rho_raw	Spearman $\hat{\rho}$ $\hat{\rho}$ UNADJUSTED â€” no confounder block DERIVED
GENE SIGNED $\hat{\rho}^1 \in \{1\}$ 81%	

w_ $\hat{\rho}$ 1
supporting â€” read these when the core raises a question

1 columns $\hat{\rho}$ coverage **100%â€”100%** (median 100%)
1x real $\hat{\rho} \notin 0$

w_max	The MAXIMUM curated evidence weight w over the gene's regulators (gene_atlas : nanmax(w)), where w comes from the same PMID-deduped ledger that feeds lit_ and fame_ . AUTHORED
GENE REAL $\hat{\rho} \notin 0$ 100%	

LIT_ $\hat{\rho}$ 7
supporting â€” read these when the core raises a question

7 columns $\hat{\rho}$ coverage **21%â€”21%** (median 21%)
4x count $\hat{\rho}$ 2x boolean $\hat{\rho}$ 1x identifier

lit_agrees_identity	Does identity name the same family as the most-published one for this gene â€” True on 100 of 323 (31%). AUTHORED
GENE BOOLEAN 21%	
lit_agrees_magnitude	Does the model's largest- $\hat{\rho}^2$ family match the most-published one (96 of 329 genes, 29%)? AUTHORED
GENE BOOLEAN 21%	

lit_family GENE IDENTIFIER 21%	The most-published seed family for this gene by distinct low-throughput functional PMIDs. AUTHORED
lit_margin GENE COUNT 21%	Gap between the top and second literature family, in PMIDs â€” median 1.00 , i.e. the literature's 'winner' is usually decided by a SINGLE paper. â€” gate on this before reading lit_agrees_* : a 1-PMID margin is a coin flip dressed as ground truth. AUTHORED Defined on: genes with a versioned literature canonical family (329 of 1,549, sha256-stamped, MH-196)
lit_n_lit_families GENE COUNT 21%	â€”â€” A STUDY-COUNT AXIS, NOT A BIOLOGICAL GROUND TRUTH â€” read the name with suspicion. eval/lit_ground_truth.py defines the 'canonical' family of a gene as the argmax of DISTINCT PubMed IDs carrying low-throughput functional evidence (reporter / western / proteomics; miRTarBase-Weak and TarBase-Positive excluded). AUTHORED
lit_n_pmids GENE COUNT 21%	Distinct low-throughput functional PMIDs behind lit_family. Median 3 â€” the 'ground truth' rests on a handful of papers per gene. AUTHORED Defined on: genes with a versioned literature canonical family (329 of 1,549, sha256-stamped, MH-196)
lit_tier GENE COUNT 21%	Literature-evidence tier for the gene (levels 2, 3 and 4; 4 is the most populated at 132 genes). AUTHORED

CPTAC_ 24

reference â€” context, provenance and rarely-quoted detail

CPTAC protein channel. 24 columns coverage 42%â€”89% (median 66%) 18 share the block description 18x signed 1 2x boolean 2x real, signed 2x count	
cptac_prosp_abund_rho_disc GENE SIGNED 1 AGG ARM 72%	CPTAC prospective cohort (n% ^101) unweighted abundance-sum reference Spearman against the discovery layer DERIVED Defined on: genes/edges present in the CPTAC cohort
cptac_prosp_abund_rho_prot GENE SIGNED 1 AGG ARM 73%	CPTAC prospective cohort (n% ^101) unweighted abundance-sum reference Spearman against protein DERIVED Defined on: genes/edges present in the CPTAC cohort
cptac_prosp_abund_rho_rna GENE SIGNED 1 AGG ARM 74%	CPTAC prospective cohort (n% ^101) unweighted abundance-sum reference Spearman against mRNA DERIVED Defined on: genes/edges present in the CPTAC cohort
cptac_prosp_agg_beats_abund_prot GENE BOOLEAN 73%	Does the learned 2-weighted aggregate couple more negatively to PROTEIN than a plain abundance sum, in the CPTAC prospective cohort? AUTHORED
cptac_prosp_agg_ret_prot GENE REAL, SIGNED 59%	rho_prot(composition-adjusted) / rho_prot_raw â€” the ADJUSTMENT sense of retention, on the prospective cohort. â€”... The denominator is ALREADY GATED at RHO_GATE in cptac_card.py:322, so values outside [0,1] are NOT the vanishing-denominator artifact of axiom 5 (verified: 0 rows with raw < 0.02). AUTHORED
cptac_prosp_agg_rho_disc GENE SIGNED 1 AGG ARM 72%	CPTAC prospective cohort (n% ^101) 2-weighted aggregate Spearman against the discovery layer DERIVED Defined on: genes/edges present in the CPTAC cohort
cptac_prosp_agg_rho_disc_raw GENE SIGNED 1 72%	CPTAC prospective cohort (n% ^101) 2-weighted aggregate Spearman against the discovery layer UNADJUSTED â€” no confounder block DERIVED Defined on: genes/edges present in the CPTAC cohort

cptac_prosp_agg_rho_prot GENE SIGNED $\hat{A}^{\sim 1} \hat{A} \epsilon 1$ AGG ARM 73%	CPTAC prospective cohort ($n \approx 101$) $\hat{A}$ $\hat{I}^2$ -weighted aggregate $\hat{A}$ $\cdot$ Spearman $\hat{I}$ $\hat{A}$ $\cdot$ against protein DERIVED <i>Defined on: genes/edges present in the CPTAC cohort</i>
cptac_prosp_agg_rho_prot_raw GENE SIGNED $\hat{A}^{\sim 1} \hat{A} \epsilon 1$ 73%	CPTAC prospective cohort ($n \approx 101$) $\hat{A}$ $\hat{I}^2$ -weighted aggregate $\hat{A}$ $\cdot$ Spearman $\hat{I}$ $\hat{A}$ $\cdot$ against protein $\hat{A}$ $\cdot$ UNADJUSTED $\hat{A} \epsilon$ no confounder block DERIVED <i>Defined on: genes/edges present in the CPTAC cohort</i>
cptac_prosp_agg_rho_rna GENE SIGNED $\hat{A}^{\sim 1} \hat{A} \epsilon 1$ AGG ARM 74%	CPTAC prospective cohort ($n \approx 101$) $\hat{A}$ $\hat{I}^2$ -weighted aggregate $\hat{A}$ $\cdot$ Spearman $\hat{I}$ $\hat{A}$ $\cdot$ against mRNA DERIVED <i>Defined on: genes/edges present in the CPTAC cohort</i>
cptac_prosp_agg_rho_rna_raw GENE SIGNED $\hat{A}^{\sim 1} \hat{A} \epsilon 1$ 74%	CPTAC prospective cohort ($n \approx 101$) $\hat{A}$ $\hat{I}^2$ -weighted aggregate $\hat{A}$ $\cdot$ Spearman $\hat{I}$ $\hat{A}$ $\cdot$ against mRNA $\hat{A}$ $\cdot$ UNADJUSTED $\hat{A} \epsilon$ no confounder block DERIVED <i>Defined on: genes/edges present in the CPTAC cohort</i>
cptac_prosp_n_arms GENE COUNT 89%	Arms contributing to this gene's prospective-cohort aggregate (median 2, max 61). AUTHORED
cptac_t105_abund_rho_disc GENE SIGNED $\hat{A}^{\sim 1} \hat{A} \epsilon 1$ 60%	CPTAC TCGA-overlap cohort (n=105) $\hat{A}$ $\cdot$ unweighted abundance-sum reference $\hat{A}$ $\cdot$ Spearman $\hat{I}$ $\hat{A}$ $\cdot$ against the discovery layer DERIVED <i>Defined on: genes/edges present in the CPTAC cohort</i>
cptac_t105_abund_rho_prot GENE SIGNED $\hat{A}^{\sim 1} \hat{A} \epsilon 1$ 60%	CPTAC TCGA-overlap cohort (n=105) $\hat{A}$ $\cdot$ unweighted abundance-sum reference $\hat{A}$ $\cdot$ Spearman $\hat{I}$ $\hat{A}$ $\cdot$ against protein DERIVED <i>Defined on: genes/edges present in the CPTAC cohort</i>
cptac_t105_abund_rho_rna GENE SIGNED $\hat{A}^{\sim 1} \hat{A} \epsilon 1$ 60%	CPTAC TCGA-overlap cohort (n=105) $\hat{A}$ $\cdot$ unweighted abundance-sum reference $\hat{A}$ $\cdot$ Spearman $\hat{I}$ $\hat{A}$ $\cdot$ against mRNA DERIVED <i>Defined on: genes/edges present in the CPTAC cohort</i>
cptac_t105_agg_beats_abund_prot GENE BOOLEAN 60%	As the prospective twin, on the TCGA-105 overlap cohort. AUTHORED
cptac_t105_agg_ret_prot GENE REAL, SIGNED 42%	As the prospective twin, TCGA-105. AUTHORED
cptac_t105_agg_rho_disc GENE SIGNED $\hat{A}^{\sim 1} \hat{A} \epsilon 1$ AGG ARM 60%	CPTAC TCGA-overlap cohort (n=105) $\hat{A}$ $\hat{I}^2$ -weighted aggregate $\hat{A}$ $\cdot$ Spearman $\hat{I}$ $\hat{A}$ $\cdot$ against the discovery layer DERIVED <i>Defined on: genes/edges present in the CPTAC cohort</i>
cptac_t105_agg_rho_disc_raw GENE SIGNED $\hat{A}^{\sim 1} \hat{A} \epsilon 1$ 60%	CPTAC TCGA-overlap cohort (n=105) $\hat{A}$ $\hat{I}^2$ -weighted aggregate $\hat{A}$ $\cdot$ Spearman $\hat{I}$ $\hat{A}$ $\cdot$ against the discovery layer $\hat{A}$ $\cdot$ UNADJUSTED $\hat{A} \epsilon$ no confounder block DERIVED <i>Defined on: genes/edges present in the CPTAC cohort</i>
cptac_t105_agg_rho_prot GENE SIGNED $\hat{A}^{\sim 1} \hat{A} \epsilon 1$ AGG ARM 60%	CPTAC TCGA-overlap cohort (n=105) $\hat{A}$ $\hat{I}^2$ -weighted aggregate $\hat{A}$ $\cdot$ Spearman $\hat{I}$ $\hat{A}$ $\cdot$ against protein DERIVED <i>Defined on: genes/edges present in the CPTAC cohort</i>
cptac_t105_agg_rho_prot_raw GENE SIGNED $\hat{A}^{\sim 1} \hat{A} \epsilon 1$ 60%	CPTAC TCGA-overlap cohort (n=105) $\hat{A}$ $\hat{I}^2$ -weighted aggregate $\hat{A}$ $\cdot$ Spearman $\hat{I}$ $\hat{A}$ $\cdot$ against protein $\hat{A}$ $\cdot$ UNADJUSTED $\hat{A} \epsilon$ no confounder block DERIVED <i>Defined on: genes/edges present in the CPTAC cohort</i>
cptac_t105_agg_rho_rna GENE SIGNED $\hat{A}^{\sim 1} \hat{A} \epsilon 1$ AGG ARM 60%	CPTAC TCGA-overlap cohort (n=105) $\hat{A}$ $\hat{I}^2$ -weighted aggregate $\hat{A}$ $\cdot$ Spearman $\hat{I}$ $\hat{A}$ $\cdot$ against mRNA DERIVED <i>Defined on: genes/edges present in the CPTAC cohort</i>

cptac_t105_agg_rho_rna_raw GENE SIGNED 1 60%	CPTAC TCGA-overlap cohort (n=105) $\hat{A} \cdot \hat{I}^2$ -weighted aggregate $\hat{A} \cdot \text{Spearman } \hat{I} \hat{A} \cdot$ against mRNA $\hat{A} \cdot$ UNADJUSTED $\hat{A} \cdot$ no confounder block DERIVED Defined on: genes/edges present in the CPTAC cohort
cptac_t105_n_arms GENE COUNT 85%	Arms behind the TCGA-105 aggregate (median 2, max 57). See the prospective twin. AUTHORED Defined on: genes/edges present in the CPTAC cohort

FRAC_ 3
reference $\hat{A} \cdot$ context, provenance and rarely-quoted detail

3 columns $\hat{A} \cdot$ coverage 60%$\hat{A} \cdot$100% (median 82%) 3$\times$ fraction 0$\hat{A} \cdot$1	
frac_edges_realized GENE FRACTION 0 82%	Fraction of the gene's scored edges that realize a coupling below the threshold. AUTHORED
frac_gain_true_acquired GENE FRACTION 0 60%	Fraction of the gene's acquired-repression calls that survive the healthy-leg triage $\hat{A} \cdot$ i.e. are not collapse_blind artifacts. AUTHORED
frac_identified GENE FRACTION 0 100%	Fraction of this gene's families reaching $ z > 2$ on the UNCALIBRATED SD. AUTHORED

DOMINANT_ 4
reference $\hat{A} \cdot$ context, provenance and rarely-quoted detail

The top-ranked regulator of this gene in one state (HLY / NAT / TUM), by GLOBAL dose rank. The inputs to regulatory_handoff . 4 columns $\hat{A} \cdot$ coverage 77%$\hat{A} \cdot$83% (median 82%) $\hat{A} \cdot$ 3 share the block description 3$\times$ identifier $\hat{A} \cdot$ 1$\times$ categorical	
dominant_HLY GENE IDENTIFIER 77%	in the GTEx-healthy state DERIVED
dominant_NAT GENE IDENTIFIER 81%	in matched normal (NAT) DERIVED
dominant_TUM GENE IDENTIFIER 82%	The arm carrying the largest share of the gene's tumour pressure budget (miR-21-5p on 56 genes, miR-200c-3p next). AUTHORED
dominant_edge_shift_class GENE CATEGORICAL 83%	per edge $\hat{A} \cdot$ shift $\hat{A} \cdot$ class label DERIVED

STATIC_ 1
reference $\hat{A} \cdot$ context, provenance and rarely-quoted detail

1 columns $\hat{A} \cdot$ coverage 87%$\hat{A} \cdot$87% (median 87%) 1$\times$ identifier	
static_owner_family GENE IDENTIFIER 87%	The static (dose-based) owner of the gene, for comparison against the realization owner. AUTHORED Defined on: MULTI-FAMILY genes only (an owner needs >1 family to choose between)

TOTAL_ 1

reference “ context, provenance and rarely-quoted detail

1 columns coverage 100%100% (median 100%)
1x real 0

total_pressure

Sum over the gene's regulators. AUTHORED

GENE REAL 0 100%

OWNER_ 1

reference “ context, provenance and rarely-quoted detail

1 columns coverage 41%41% (median 41%)
1x boolean

owner_agrees

Whether the realization owner and the static owner agree. Defined only where both exist. AUTHORED

GENE BOOLEAN 41%

| Defined on: MULTI-FAMILY genes only (an owner needs >1 family to choose between)

MEDIAN_ 1

reference “ context, provenance and rarely-quoted detail

1 columns coverage 100%100% (median 100%)
1x real 0

median_retention

A per-row median of a lower-rung quantity; check agg_of . AUTHORED

GENE REAL 0 100%

ARMRES_ 7

reference “ context, provenance and rarely-quoted detail

IS THIS GENE MODELABLE AT ARM RESOLUTION “ the gene-level roll-up of arm identifiability
(card_ladders.gene_arm_resolution , 2026-08-19).
7 columns coverage 18%18% (median 18%) 4 share the block description
2x signed 11 2x count 1x categorical 1x fraction 1 1x real 0

armres_best_drho

the best DERIVED

GENE SIGNED ^11 18%

armres_class

arm_modelable (every multi-arm cell resolvable, 68 genes) partial (37) family_only (no cell resolvable, 168). Absent = the gene has no multi-arm cell at all, so the question is undefined. AUTHORED

GENE CATEGORICAL 18%

armres_frac_resolvable

Fraction of this gene's multi-arm cells where the arms are separable. AUTHORED

GENE FRACTION 1 18%

armres_med_drho

Median of_drho over the gene's multi-arm cells “ negative means arm resolution predicts repression better out of fold. AUTHORED

GENE SIGNED ^11 18%

armres_med_sep_z

median z-score DERIVED

GENE REAL 0 18%

armres_n_arms_split

count

arms

DERIVED

GENE

COUNT

18%

armres_n_multi_cells

count

DERIVED

GENE

COUNT

18%

IDENTITY_ 1

reference context, provenance and rarely-quoted detail

1 columns coverage 87%87% (median 87%)

1x boolean

identity_eq_magnitude

Do the identity and magnitude answers name the SAME top family. Exactly 100% agreement at n_fam==1 by construction; 76% at n_fam>=2. Read it only on multi-family genes.

GENE

BOOLEAN

AGG FAMILY

87%

AUTHORED

HEUR_ 5

reference context, provenance and rarely-quoted detail

5 columns coverage 89%91% (median 91%)

2x signed 1 1 1x count 1x boolean 1x categorical

heur_delta_tumor_nat

Tumour-minus-NAT change in the heuristic pressure correlation.

GENE

SIGNED 1 1

89%

AUTHORED

heur_n_regulators

A DIFFERENT PIPELINE'S COUNT not the model's. It is edges.groupby('gene')['miRNA'].nunique() inside analyses/misc/mirna_comovement.py, i.e.

GENE

COUNT

91%

AUTHORED

heur_net_repressed_tumor

Is the gene net-repressed in tumour, per the RETIRED heuristic lane. True for 282 of 1,409 genes.

GENE

BOOLEAN

91%

AUTHORED

heur_repression_class

Cross-state repression class from the RETIRED heuristic lane (never_repressed / lost_repression / gained_repression / constitutive_repressed). 867 of 1,409 genes are never_repressed.

GENE

CATEGORICAL

91%

AUTHORED

heur_rho_pressure_tumor

Spearman of the gene's HEURISTIC aggregate pressure against its mRNA in tumour.

GENE

SIGNED 1 1

91%

AUTHORED

GREAL_ 10

reference context, provenance and rarely-quoted detail

Realization ladder for the GENE the mirror of real_ : how many of THIS gene's regulators realize at each calibrated-coupling depth (c05/c10/c15/c20/c30). ... Cross-rung control: the gene-side totals reproduce the arm-side exactly.

10 columns coverage 29%83% (median 83%) 8 share the block description

7x count 2x signed 1 1 1x fraction 01

greal_best_coupling

the best

DERIVED

GENE

SIGNED 1 1

83%

Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) the GENE-side ladder; rates gated n3

greal_frac_c10 GENE FRACTION 0 29%	Gene-rung twin of real_frac_c10 defined on 446 genes against greal_n_c10's 1,289; gated-out genes have a median greal_n_scored of 1. Exact on 100%. AUTHORED Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) the GENE-side ladder; rates gated n3
greal_med_coupling GENE SIGNED 1 83%	median DERIVED Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) the GENE-side ladder; rates gated n3
greal_n_c05 GENE COUNT 83%	count at the 0.05 threshold DERIVED Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) the GENE-side ladder; rates gated n3
greal_n_c10 GENE COUNT 83%	count at the 0.10 threshold DERIVED Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) the GENE-side ladder; rates gated n3
greal_n_c15 GENE COUNT 83%	count at the 0.15 threshold DERIVED Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) the GENE-side ladder; rates gated n3
greal_n_c20 GENE COUNT 83%	count at the 0.20 threshold DERIVED Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) the GENE-side ladder; rates gated n3
greal_n_c30 GENE COUNT 83%	count at the 0.30 threshold DERIVED Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) the GENE-side ladder; rates gated n3
greal_n_scored GENE COUNT 83%	Genes/edges scored for realized coupling (median 2, max 59) the DENOMINATOR of every greal_frac_*. The greal_n_c05, n_c30 ladder is properly nested: 0 monotonicity violations and 0 rows where a threshold count exceeds n_scored (verified). AUTHORED Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) the GENE-side ladder; rates gated n3
greal_n_sig GENE COUNT 83%	count DERIVED Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) the GENE-side ladder; rates gated n3

ACQUIRED_ 2

reference context, provenance and rarely-quoted detail

Acquisition of regulatory dose relative to a healthy reference (GTEx or NAT).

2 columns coverage 91%91% (median 91%) 2 share the block description

2x real, signed

acquired_vs_gtex GENE REAL, SIGNED 91%	GTEx DERIVED
acquired_vs_nat GENE REAL, SIGNED 91%	in matched normal (NAT) DERIVED

PRESSURE_ 1

reference " context, provenance and rarely-quoted detail

1 columns coverage 91%91% (median 91%)

1x real 0

pressure_tumor Aggregate incoming pressure on the gene in the named state. AUTHORED

GENE

REAL 0

91%

REALIZATION_ 1

reference " context, provenance and rarely-quoted detail

1 columns coverage 42%42% (median 42%)

1x identifier

realization_owner Within-patient realization quantities " the score, its identity allocation, and who owns it. AUTHORED

GENE

IDENTIFIER

42%

Defined on: MULTI-FAMILY genes only (an owner needs >1 family to choose between)

TCGA_ 4

reference " context, provenance and rarely-quoted detail

The TCGA mRNA leg carried beside the CPTAC columns so the two are comparable on the same gene rows.

4 columns coverage 89%89% (median 89%) 3 share the block description

3x signed 1 1 1x count

tcga_abund_rho_rna unweighted abundance-sum reference Spearman against mRNA DERIVED

GENE

SIGNED 1 1

AGG ARM

89%

tcga_agg_rho_rna Gene-level 2-weighted aggregate coupling to mRNA in TCGA (median 0.05) against tcga_abund_rho_rna 's 0.02 " the abundance-sum reference. AUTHORED

GENE

SIGNED 1 1

AGG ARM

89%

tcga_agg_rho_rna_raw 2-weighted aggregate Spearman against mRNA UNADJUSTED " no confounder block DERIVED

GENE

SIGNED 1 1

89%

tcga_n_arms count arms DERIVED

GENE

COUNT

89%

DISC_ 2

reference " context, provenance and rarely-quoted detail

2 columns coverage 3%3%100% (median 51%)

1x categorical 1x count

disc_gold_families Which seed families this row's discovery-queue edges belong to, semicolon-separated. Usually ONE at the arm rung (an arm has one seed). AUTHORED

GENE

CATEGORICAL

3%

disc_n_gold_edges

How many of the 157 discovery-queue edges name this gene (max 4).

AUTHORED

GENE

COUNT

100%

MAX_ 1
reference context, provenance and rarely-quoted detail

1 columns

coverage

100%100%

(median 100%)

1x fraction

01

max_beta_frac_sd

Maximum over the row's constituent units.

AUTHORED

GENE

FRACTION 01

100%

REGULATORY_ 1
reference context, provenance and rarely-quoted detail

1 columns

coverage

100%100%

(median 100%)

1x boolean

regulatory_handoff

Does the gene's globally-dominant regulator differ between healthy and tumour OR between NAT and tumour.

AUTHORED

GENE

BOOLEAN

100%

MEAN_ 1
reference context, provenance and rarely-quoted detail

1 columns

coverage

82%82%

(median 82%)

1x real, signed

mean_dTarget

A per-row mean of a lower-rung quantity; check agg_of for which rung it summarises.

AUTHORED

GENE

REAL, SIGNED

82%

CONCENTRATION_ 1
reference context, provenance and rarely-quoted detail

1 columns

coverage

53%53%

(median 53%)

1x fraction

01

concentration_adj

$(\text{concentration}^{1/n_{fam}} / (1^{1/n_{fam}}))^{[0,1]}$ how concentrated the gene is RELATIVE to the most diffuse its own design allows.

AUTHORED

GENE

FRACTION 01

53%

FAM_ 1
reference context, provenance and rarely-quoted detail

1 columns

coverage

81%81%

(median 81%)

1x signed

11

fam_pooled_rho_adj

Seed-family structure the arm sits in membership, affinity and dominance within it.

AUTHORED

GENE

SIGNED 11

81%

gene_family_card.tsv

gene_family

One row per one (seed family, gene) pair. Keyed [gene, family] it cannot express a property of the family itself. That is the seed_family card's job.

61 columns 5,117 rows 12 blocks median coverage 97%

Contents 12 blocks, most important first

1. (bare) 8 CORE

2. n_3 SUPPORTING

3. ctx_16 SUPPORTING

4. beta_7 SUPPORTING

5. comp_16 REFERENCE

6. identity_4 REFERENCE
7. pip_2 REFERENCE

8. w_1 REFERENCE

9. m_1 REFERENCE

10. composition_1 REFERENCE

11. prior_1 REFERENCE

12. retention_1 REFERENCE

(BARE) 8

core the columns findings rest on

8 columns coverage 95%100% (median 100%)
3x identifier 3x real 0 1x boolean 1x real, signed

arms FAMILY IDENTIFIER 100%	The arms collapsed into this (gene, family) cell, semicolon-separated (hsa-miR-17-5p; hsa-miR-20a-5p). AUTHORED
beta FAMILY REAL 0 100%	FAMILY-rung coupling coefficient readouts.run(level='family') , the same-seed collapse APPLIED. AUTHORED
family KEY IDENTIFIER 100%	KEY seed family of the regulator, as used by the family-rung fit. AUTHORED
gene KEY IDENTIFIER 100%	KEY HGNC gene symbol. AUTHORED
identified FAMILY BOOLEAN 100%	z > 2 on the UNCALIBRATED SD 27.1% of cells. cal_identified is the honest version. AUTHORED
identity FAMILY REAL, SIGNED 95%	Shapley/LMG identity at the FAMILY rung same estimator as the edge card's identity , different unit. AUTHORED
retention FAMILY REAL 0 100%	beta_deconv / beta_core at the family grain the ADJUSTMENT sense of retention (what fraction of the coupling survives the composition block), never the patient-baseline sense. AUTHORED
z FAMILY REAL 0 100%	beta / beta_sd for the family cell, on the UNCALIBRATED posterior SD. AUTHORED

N_ 3

supporting “ read these when the core raises a question

3 columns · coverage 100%“100% (median 100%) 2× count · 1× constant		
n_arms	How many arms of THIS family sit in THIS gene's design. æ... FAMILY rung, VERIFIED (2026-08-19): it varies within gene for 241 of 1,549 genes, and the per-gene SUM of these equals the gene card's n_arms exactly æ” a clean cross-rung consistency check. AUTHORED	
n_fam	How many families compete at this gene. beta is mathematically INERT at 1. AUTHORED	
n_samples	Samples the family cell was fit on æ” CONSTANT 1,040 across the card. AUTHORED	

CTX_ 16

supporting “ read these when the core raises a question

Gene-design context from the decoy/atlas layer (learned/analyses/gene_atlas.py via card_context): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it. 16 columns · coverage 81%æ”100% (median 96%) · 13 share the block description 4× signed æ^1æ 1 · 3× real æ%æ 0 · 3× real, signed · 3× count · 1× categorical · 1× fraction 0æ”1 · 1× boolean		
ctx_apriori_class	class label	DERIVED
ctx_ceiling	The UPPER BOUND on what ANY estimator can achieve for this gene æ” the 5-fold out-of-fold R² of an UNRESTRICTED OLS on the gene's real design (card_context.py:15). AUTHORED	
ctx_collapse	n_arm / n_fam for the gene æ” how much curation BUNDLES same-seed mates onto it (1æ”5). AUTHORED	
ctx_d_collin	change	DERIVED
ctx_d_fam	change · per seed family	DERIVED
ctx_dose_max	dose · maximum	DERIVED
ctx_dose_med	dose · median	DERIVED
ctx_frac_abund	fraction · unweighted abundance-sum reference	DERIVED
ctx_gap_core	gap · core confounder block	DERIVED
ctx_gap_d_dose	gap · change · dose	DERIVED

ctx_gap_deconv	gap $\hat{\cdot}$ composition-adjusted (deconvolution block)	DERIVED
GENE	SIGNED $\hat{\cdot}$ 1	95%
ctx_gap_retention	gap $\hat{\cdot}$ retention	DERIVED
GENE	REAL, SIGNED	92%
ctx_measurable	Is the gene's ctx_ceiling above CEILING_ROBUST (0.02, not 0 $\hat{\cdot}$ MH-144)?	AUTHORED
GENE	BOOLEAN	99%
ctx_n_abund	count $\hat{\cdot}$ unweighted abundance-sum reference	DERIVED
GENE	COUNT	96%
ctx_n_fam_abund	count $\hat{\cdot}$ per seed family $\hat{\cdot}$ unweighted abundance-sum reference	DERIVED
GENE	COUNT	96%
ctx_n_fam_multi	count $\hat{\cdot}$ per seed family	DERIVED
GENE	COUNT	96%

BETA_ $\hat{\cdot}$ 7
supporting $\hat{\cdot}$ read these when the core raises a question

Posterior summaries of the coupling coefficient from the dense Gibbs fit (learned/readouts.py). `_sd` is the posterior SD, `_frac` the share of the gene's total beta.

7 columns $\hat{\cdot}$ coverage **100%** (median 100%) $\hat{\cdot}$ **7** share the block description
4x fraction 0 $\hat{\cdot}$ 1 $\hat{\cdot}$ 3x real $\hat{\cdot}$ 0

beta_deconv	composition-adjusted (deconvolution block)	DERIVED
FAMILY	REAL $\hat{\cdot}$ 0	100%
beta_frac	fraction	DERIVED
FAMILY	FRACTION 0 $\hat{\cdot}$ 1	100%
beta_frac_abs	fraction	DERIVED
FAMILY	FRACTION 0 $\hat{\cdot}$ 1	100%
beta_frac_deconv	fraction $\hat{\cdot}$ composition-adjusted (deconvolution block)	DERIVED
FAMILY	FRACTION 0 $\hat{\cdot}$ 1	100%
beta_frac_sd	fraction $\hat{\cdot}$ standard deviation	DERIVED
FAMILY	FRACTION 0 $\hat{\cdot}$ 1	100%
beta_sd	standard deviation	DERIVED
FAMILY	REAL $\hat{\cdot}$ 0	100%
beta_sd_deconv	standard deviation $\hat{\cdot}$ composition-adjusted (deconvolution block)	DERIVED
FAMILY	REAL $\hat{\cdot}$ 0	100%

COMP_ 16

reference “ context, provenance and rarely-quoted detail

Cell-composition drivers of this gene's coupling (compartment_*): which deconvolved fraction explains the correlation, its retention under adjustment, and its share.

16 columns coverage 66%“97% (median 81%) 7 share the block description
6× categorical 6× signed 1 3× real, signed 1× real 0

comp_cptac_prot_driver	CPTAC against protein DERIVED
GENE CATEGORICAL 66%	Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)
comp_cptac_prot_driver_ret	1 ^' comp_cptac_prot_driver_share , exactly. Median +0.398 raw, +0.469 gated to Ĭ_raw % 0.10 “ quote the gated value. See the mRNA twin. AUTHORED
GENE REAL, SIGNED 66%	Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)
comp_cptac_prot_driver_share	Protein-layer twin of comp_tcga_mrna_driver_share . Same exact complementarity with _driver_ret (8.9e-16 on 918 rows). AUTHORED
GENE REAL, SIGNED 66%	
comp_cptac_prot_rho_raw	Unadjusted aggregate PROTEIN coupling “ denominator of comp_cptac_prot_driver_ret . AUTHORED
GENE SIGNED ^'1 81%	
comp_cptac_prot_top_budget	CPTAC against protein the top-ranked pressure budget DERIVED
GENE CATEGORICAL 81%	Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)
comp_cptac_prot_top_budget_r	CPTAC against protein the top-ranked pressure budget correlation DERIVED
GENE SIGNED ^'1 81%	Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)
comp_cptac_prot_top_target	CPTAC against protein the top-ranked DERIVED
GENE CATEGORICAL 81%	Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)
comp_cptac_prot_top_target_r	CPTAC against protein the top-ranked correlation DERIVED
GENE SIGNED ^'1 81%	Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)
comp_tcga_mrna_driver	WHICH compartment drives the gene's mRNA coupling “ the one whose removal moves it most (CAFs 279 purity 174 prolif 148 Endothelial 122 Myeloid 109). AUTHORED
GENE CATEGORICAL 81%	
comp_tcga_mrna_driver_ret	1 ^' comp_tcga_mrna_driver_share , exactly (see there). The RETENTION orientation: what fraction of the coupling survives removing the driver compartment. AUTHORED
GENE REAL, SIGNED 81%	
comp_tcga_mrna_driver_share	Share of the gene's mRNA coupling attributable to the single compartment that most changes it “ axiom 8's composition GATE, the axis that separates the harm tails at p=0.00999 where the ensemble axis gives p=0.10. AUTHORED
GENE REAL 0 81%	
comp_tcga_mrna_rho_raw	The gene's UNADJUSTED aggregate mRNA coupling “ the DENOMINATOR of comp_tcga_mrna_driver_ret . AUTHORED
GENE SIGNED ^'1 97%	
comp_tcga_mrna_top_budget	The compartment the gene's miRNA BUDGET loads on most “ a property of the REGULATORS. AUTHORED
GENE CATEGORICAL 97%	
comp_tcga_mrna_top_budget_r	TCGA the top-ranked pressure budget correlation DERIVED
GENE SIGNED ^'1 97%	

comp_tcga_mrna_top_target

The compartment the TARGET GENE's expression loads on most.

AUTHORED

GENE

CATEGORICAL

97%

comp_tcga_mrna_top_target_r

TCGA · the top-ranked · correlation

DERIVED

GENE

SIGNED · 1

97%

IDENTITY_ · 4

reference · context, provenance and rarely-quoted detail

4 columns · coverage 95%–97% (median 95%)
2× fraction 0–1 · 1× real, signed · 1× boolean

identity_abs

|identity| / Î|identity| per gene · the BOUNDED companion to identity , always in [0,1] (verified: 0 non-NaN values outside it on either card).

AUTHORED

FAMILY

FRACTION 0–1

95%

identity_coherence

1 / Î|identity| per GENE, in (0,1] · identity's own sign coherence. 1.0 when nothing cancels, · 0 as suppressor cancellation grows.

AUTHORED

FAMILY

FRACTION 0–1

95%

identity_deconv

Shapley/LMG attribution on R^2 with bagged-NNLS weights · WHO regulates the gene, collinearity-fair.

AUTHORED

FAMILY

REAL, SIGNED

97%

identity_reliable

identity_coherence ≥ 0.5 AND |identity| ≤ 1 . Admits 93.3% of edge rows and 92.6% of gene_family rows, and excludes every row whose |identity| exceeds 1 (119/119 and 118/118).

AUTHORED

FAMILY

BOOLEAN

95%

PIP_ · 2

reference · context, provenance and rarely-quoted detail

2 columns · coverage 100%–100% (median 100%)
1× constant · 1× fraction 0–1

pip_dense

· CONSTANT by construction (Î%|1 dense readout).

AUTHORED

FAMILY

CONSTANT

100%

pip_discovery

Posterior inclusion probability. pip_dense comes from the pi=1 coupling readout; pip_discovery from the evidence-pi readout.

AUTHORED

FAMILY

FRACTION 0–1

100%

W_ · 1

reference · context, provenance and rarely-quoted detail

1 columns · coverage 100%–100% (median 100%)
1× real ≥ 0

w_max

Maximum curated EVIDENCE weight for this (gene, family) cell. Family-rung here; gene-rung on the gene card.

AUTHORED

GENE

REAL ≥ 0

100%

M_ 1

reference " context, provenance and rarely-quoted detail

1 columns coverage 100%100% (median 100%)

1x real 0

m_nnls Bagged-NNLS weight for this (gene, family) cell. AUTHORED

FAMILY

REAL 0

100%

COMPOSITION_ 1

reference " context, provenance and rarely-quoted detail

1 columns coverage 100%100% (median 100%)

1x categorical

composition_class Composition-adjustment verdict for this row (e.g. composition_explained when the effect does not survive the deconvolution block). AUTHORED

FAMILY

CATEGORICAL

100%

PRIOR_ 1

reference " context, provenance and rarely-quoted detail

1 columns coverage 100%100% (median 100%)

1x fraction 1

prior_pi The evidence prior fed to the discovery readout. AUTHORED

FAMILY

FRACTION 1

100%

RETENTION_ 1

reference " context, provenance and rarely-quoted detail

1 columns coverage 100%100% (median 100%)

1x boolean

retention_reliable Fraction of an effect surviving covariate adjustment (adjusted / raw). AUTHORED

FAMILY

BOOLEAN

100%

arm

296 columns **2,450** rows **43** blocks median coverage **30%**

1. adm_ 4 Â· CORE	22. seq_ 4 Â· REFERENCE
2. arm_ 7 Â· CORE	23. sub_ 6 Â· REFERENCE
3. seed_ 1 Â· CORE	24. cov_ 22 Â· REFERENCE
4. healthy_ 3 Â· CORE	25. cur_ 2 Â· REFERENCE
5. site_ 12 Â· SUPPORTING	26. iso_ 5 Â· REFERENCE
6. arb_ 16 Â· SUPPORTING	27. field_ 5 Â· REFERENCE
7. (bare) 4 Â· SUPPORTING	28. famrole_ 6 Â· REFERENCE
8. fame_ 28 Â· SUPPORTING	29. bc_ 11 Â· REFERENCE
9. ago_ 4 Â· SUPPORTING	30. gctx_ 16 Â· REFERENCE
10. hly_ 7 Â· SUPPORTING	31. surv_ 8 Â· REFERENCE
11. dose_ 3 Â· SUPPORTING	32. aid_ 4 Â· REFERENCE
12. abund_ 5 Â· SUPPORTING	33. tier_ 5 Â· REFERENCE
13. attr_ 14 Â· REFERENCE	34. wshift_ 6 Â· REFERENCE
14. ctx_ 2 Â· REFERENCE	35. surrogate_ 2 Â· REFERENCE
15. grank_ 3 Â· REFERENCE	36. disc_ 2 Â· REFERENCE
16. real_ 11 Â· REFERENCE	37. foc_ 7 Â· REFERENCE
17. ts_ 10 Â· REFERENCE	38. cnv_ 9 Â· REFERENCE
18. isoc_ 5 Â· REFERENCE	39. cnvc_ 5 Â· REFERENCE
19. sdsz_ 4 Â· REFERENCE	40. model_ 1 Â· REFERENCE
20. chim_ 5 Â· REFERENCE	41. comv_ 2 Â· REFERENCE
21. fst_ 16 Â· REFERENCE	42. cload_ 3 Â· REFERENCE
	43. hx_ 1 Â· REFERENCE

core “ the columns findings rest on

AUTHORED

AUTHORED

AUTHORED

adm_n_with_site

ARM

COUNT

29%

Edges of this arm whose target carries a seed site “ 89.9% of edges (4,436/4,937), so the site requirement is NOT the binding constraint; the evidence bar is (admissible drops it to 60.4%). 19.7% of arms have none.

AUTHORED

Defined on: edges/arms with an admissibility tag (4,937 edges; has_site 89.9% ^ expressed 63.9% ^ admissible 60.4%)

ARM_ ^ 7

core “ the columns findings rest on

7 columns ^ coverage 15%“20% (median 20%)

4x real, signed ^ 2x real “ 0 ^ 1x percent

arm_iqr

Interquartile range of the arm's tumour RPM “ its DYNAMIC RANGE.

AUTHORED

ARM

REAL “ 0

20%

arm_lfc_HLY_NAT_raw

log-FC GTEEx-healthy “ TCGA-NAT, raw.

AUTHORED

ARM

REAL, SIGNED

15%

arm_lfc_HLY_TUM_QN

log-FC GTEEx-healthy “ TCGA-tumour, on quantile-normalised values.

AUTHORED

ARM

REAL, SIGNED

16%

arm_lfc_HLY_TUM_raw

As arm_lfc_HLY_TUM_QN but against the RAW GTEEx median.

AUTHORED

ARM

REAL, SIGNED

15%

arm_lfc_NAT_TUM

log-FC of the arm's median abundance, matched NAT “ tumour.

AUTHORED

ARM

REAL, SIGNED

20%

arm_med_rpm

The arm's MEDIAN RPM across tumour samples “ its abundance level, gene-free. Defined on the 19.8% of arms with tumour expression data.

AUTHORED

ARM

REAL “ 0

20%

Defined on: edges present in the progression panel (arm expressed + state measured)

arm_pct_above_floor

“ THE NAME READS BACKWARDS. Computed as 100^mean(x > FLOOR) “ the percent of samples ABOVE the detection floor, i.e.

AUTHORED

ARM

PERCENT

20%

SEED_ ^ 1

core “ the columns findings rest on

1 columns ^ coverage 100%“100% (median 100%)

1x identifier

seed_family

KEY “ the seed family itself, one row per family.

AUTHORED

ARM

IDENTIFIER

100%

HEALTHY_ ^ 3

core “ the columns findings rest on

3 columns ^ coverage 16%“20% (median 16%)

1x categorical ^ 1x real “ 0 ^ 1x boolean

healthy_leg

Provenance of this arm's GTEEx healthy leg “ measured , collapse_blind or true_absent .

AUTHORED

ARM

CATEGORICAL

20%

healthy_potential ARM REAL 0 16%	The arm's IMPUTED healthy abundance â€” a repression-POTENTIAL proxy, collapse-repaired (0.27 â€” 17.9). AUTHORED
healthy_uninformative ARM BOOLEAN 16%	Is the healthy leg both collapse_blind AND high-potential â€” i.e. the one combination where an *acquired* call is genuinely unsafe? True on 143 of 3,490 edges (4%). AUTHORED

SITE_ 12
supporting â€” read these when the core raises a question

Seed-site counts in the target 3'UTR by site type (8mer / 7mer / 6mer / non-canonical) from the genome-wide scan.

12 columns coverage 29%â€”29% (median 29%) 7 share the block description
7x count 2x fraction 0â€”1 2x real, signed 1x real 0

site_frac_8mer ARM FRACTION 0â€”1 29%	8mer share of the arm's CANONICAL sites (site_n_8mer / site_n_canonical ; median 0.150). AUTHORED
site_frac_canonical ARM FRACTION 0â€”1 29%	Canonical share of the arm's TOTAL sites (site_n_canonical / site_n_total ; median 0.054) â€” a different denominator from site_frac_8mer . AUTHORED
site_n_6mer ARM COUNT AGG EDGE 29%	count 6mer sites DERIVED Defined on: arms in the scanMiR site-type table (771) â€” HALLMARK-SCOPED, 1,432 genes only
site_n_7mer ARM COUNT AGG EDGE 29%	count 7mer sites DERIVED Defined on: arms in the scanMiR site-type table (771) â€” HALLMARK-SCOPED, 1,432 genes only
site_n_8mer ARM COUNT AGG EDGE 29%	count 8mer sites DERIVED Defined on: arms in the scanMiR site-type table (771) â€” HALLMARK-SCOPED, 1,432 genes only
site_n_canonical ARM COUNT AGG EDGE 29%	count canonical sites DERIVED Defined on: arms in the scanMiR site-type table (771) â€” HALLMARK-SCOPED, 1,432 genes only
site_n_genes_canonical ARM COUNT AGG EDGE 29%	count genes canonical sites DERIVED Defined on: arms in the scanMiR site-type table (771) â€” HALLMARK-SCOPED, 1,432 genes only
site_n_noncanon ARM COUNT AGG EDGE 29%	count non-canonical sites DERIVED Defined on: arms in the scanMiR site-type table (771) â€” HALLMARK-SCOPED, 1,432 genes only
site_n_total ARM COUNT AGG EDGE 29%	count total DERIVED Defined on: arms in the scanMiR site-type table (771) â€” HALLMARK-SCOPED, 1,432 genes only
site_repression_med ARM REAL, SIGNED AGG EDGE 29%	Median predicted repression score over the arm's sites (all negative by construction; median âˆ’0.316). A PREDICTION from sequence, not a measurement â€” never cite it as evidence of observed repression. AUTHORED Defined on: arms in the scanMiR site-type table (771) â€” HALLMARK-SCOPED, 1,432 genes only
site_repression_min ARM REAL, SIGNED AGG EDGE 29%	Strongest (most negative) predicted repression among the arm's sites (median âˆ’1.964). AUTHORED
site_sites_per_gene ARM REAL 0 29%	Canonical sites per targeted gene (median 1.34) â€” site DENSITY, separating an arm with many sites on few genes from one spread thin. site_n_canonical / site_n_genes_canonical . AUTHORED Defined on: arms in the scanMiR site-type table (771) â€” HALLMARK-SCOPED, 1,432 genes only

ARB_ 16

supporting " read these when the core raises a question

Per-ARM rollup of its edge set " how many edges and genes it regulates and its mean |beta| over them.

16 columns coverage 28%30% (median 30%) 10 share the block description

12x count 2x real 1x fraction 1x signed 1

arb_frac_identified ARM FRACTION 01 AGG EDGE 30%	arb_n_identified / arb_n_edges (verified exact on 100% of rows). AUTHORED
arb_max_abs_beta ARM REAL 0 AGG EDGE 30%	Largest $\hat{I}^2$ among the arm's edges. AUTHORED
arb_max_identity ARM SIGNED 11 AGG EDGE 28%	Largest identity among this arm's edges, over reliable rows only. ... Now runs 0.255 it reached +1.000 it reached +740.0 until the inert beta_frac_reliable gate was replaced (2026-08-19). AUTHORED
arb_mean_abs_beta ARM REAL 0 AGG EDGE 30%	Mean $\hat{I}^2$ over the arm's edges. See arb_max_abs_beta for the one-edge degeneracy (9.3% of arms, where the two are equal by construction). AUTHORED Defined on: arms with a learned-$\hat{I}^2$ edge rollup (728)
arb_n_b001 ARM COUNT 30%	count at the 0.001 threshold DERIVED Defined on: arms with a learned-$\hat{I}^2$ edge rollup (728)
arb_n_b001_ident ARM COUNT 30%	count at the 0.001 threshold DERIVED Defined on: arms with a learned-$\hat{I}^2$ edge rollup (728)
arb_n_b005 ARM COUNT 30%	count at the 0.005 threshold DERIVED Defined on: arms with a learned-$\hat{I}^2$ edge rollup (728)
arb_n_b005_ident ARM COUNT 30%	count at the 0.005 threshold DERIVED Defined on: arms with a learned-$\hat{I}^2$ edge rollup (728)
arb_n_b010 ARM COUNT 30%	count at the 0.010 threshold DERIVED Defined on: arms with a learned-$\hat{I}^2$ edge rollup (728)
arb_n_b010_ident ARM COUNT 30%	count at the 0.010 threshold DERIVED Defined on: arms with a learned-$\hat{I}^2$ edge rollup (728)
arb_n_b025 ARM COUNT 30%	count at the 0.025 threshold DERIVED Defined on: arms with a learned-$\hat{I}^2$ edge rollup (728)
arb_n_b025_ident ARM COUNT 30%	count at the 0.025 threshold DERIVED Defined on: arms with a learned-$\hat{I}^2$ edge rollup (728)
arb_n_comp_explained ARM COUNT AGG EDGE 30%	count DERIVED Defined on: arms with a learned-$\hat{I}^2$ edge rollup (728)
arb_n_edges ARM COUNT AGG EDGE 30%	Edges this arm carries in the LEARNED MODEL's own readouts. AUTHORED
arb_n_identified ARM COUNT AGG EDGE 30%	count DERIVED Defined on: arms with a learned-$\hat{I}^2$ edge rollup (728)

arb_n_identity_reliable	Edges of this arm whose identity share is usable (identity_reliable).	AUTHORED
ARM	COUNT	AGG EDGE 30%

(BARE) 4
supporting read these when the core raises a question

Change in the arm's GLOBAL (cohort-wide) dose rank between two states. HLY_NAT is the FIELD delta, NAT_TUM the malignant step.

4 columns coverage 15%100% (median 17%) 3 share the block description

3x real, signed 1x identifier

arm	KEY IDENTIFIER 100%	KEY read mature miRNA arm (e.g. hsa-miR-21-5p). Normalised through arm_key() , which gates against the canonical universe and LOGS drops rather than silently dropping.	AUTHORED
dGlobal_HLY_NAT	ARM REAL, SIGNED 15%	in the GTEx-healthy state in matched normal (NAT)	DERIVED
		Defined on: arms with a GTEx-HEALTHY leg (the thinnest state many arms are multi-mapping-collapsed, MH-166)	
dGlobal_HLY_TUM	ARM REAL, SIGNED 15%	in the GTEx-healthy state in tumour	DERIVED
		Defined on: arms with a GTEx-HEALTHY leg (the thinnest state many arms are multi-mapping-collapsed, MH-166)	
dGlobal_NAT_TUM	ARM REAL, SIGNED 20%	in matched normal (NAT) in tumour	DERIVED
		Defined on: edges with a matched NAT leg (n~104 paired participants)	

FAME_ 28
supporting read these when the core raises a question

Study-attention axis for the arm distinct PMIDs and distinct curated genes.

28 columns coverage 94%96% (median 96%) 23 share the block description

27x count 1x fraction 01

fame_assay_binding_functional_mti_weak_studies	ARM COUNT 96%	by assay class functional assays miRNAtarget interaction evidence weak evidence study count	DERIVED
		Defined on: arms with miRTarBase curation / ledger PMIDs read FAME axes, not targetome (MH-208)	
fame_assay_binding_studies	ARM COUNT 96%	Studies whose assay class is BINDING (the marginal of the assay functional-class cross-tab).	AUTHORED
fame_assay_functional_mti_studies	ARM COUNT 96%	by assay class functional assays miRNAtarget interaction evidence study count	DERIVED
		Defined on: arms with miRTarBase curation / ledger PMIDs read FAME axes, not targetome (MH-208)	
fame_assay_functional_mti_weak_studies	ARM COUNT 96%	by assay class functional assays miRNAtarget interaction evidence weak evidence study count	DERIVED
		Defined on: arms with miRTarBase curation / ledger PMIDs read FAME axes, not targetome (MH-208)	
fame_assay_nonfunctional_mti_studies	ARM COUNT 96%	by assay class miRNAtarget interaction evidence study count	DERIVED
		Defined on: arms with miRTarBase curation / ledger PMIDs read FAME axes, not targetome (MH-208)	
fame_assay_other_functional_mti_studies	ARM COUNT 96%	by assay class functional assays miRNAtarget interaction evidence study count	DERIVED
		Defined on: arms with miRTarBase curation / ledger PMIDs read FAME axes, not targetome (MH-208)	

fame_assay_other_functional_mti_weak_studies ARM COUNT 96%	by assay class $\hat{A}$: functional assays $\hat{A}$: miRNA $\hat{A}$ “target interaction evidence $\hat{A}$: weak evidence $\hat{A}$: study count DERIVED Defined on: arms with miRTarBase curation / ledger PMIDs $\hat{A}$ ” $\hat{A}$ ” FAME axes, not targetome (MH-208)
fame_assay_other_nonfunctional_mti_studies ARM COUNT 96%	by assay class $\hat{A}$: miRNA $\hat{A}$ “target interaction evidence $\hat{A}$: study count DERIVED Defined on: arms with miRTarBase curation / ledger PMIDs $\hat{A}$ ” $\hat{A}$ ” FAME axes, not targetome (MH-208)
fame_assay_other_studies ARM COUNT 96%	by assay class $\hat{A}$: study count DERIVED Defined on: arms with miRTarBase curation / ledger PMIDs $\hat{A}$ ” $\hat{A}$ ” FAME axes, not targetome (MH-208)
fame_assay_protein_functional_mti_studies ARM COUNT 96%	by assay class $\hat{A}$: functional assays $\hat{A}$: miRNA $\hat{A}$ “target interaction evidence $\hat{A}$: study count DERIVED Defined on: arms with miRTarBase curation / ledger PMIDs $\hat{A}$ ” $\hat{A}$ ” FAME axes, not targetome (MH-208)
fame_assay_protein_functional_mti_weak_studies ARM COUNT 96%	by assay class $\hat{A}$: functional assays $\hat{A}$: miRNA $\hat{A}$ “target interaction evidence $\hat{A}$: weak evidence $\hat{A}$: study count DERIVED Defined on: arms with miRTarBase curation / ledger PMIDs $\hat{A}$ ” $\hat{A}$ ” FAME axes, not targetome (MH-208)
fame_assay_protein_nonfunctional_mti_studies ARM COUNT 96%	by assay class $\hat{A}$: miRNA $\hat{A}$ “target interaction evidence $\hat{A}$: study count DERIVED Defined on: arms with miRTarBase curation / ledger PMIDs $\hat{A}$ ” $\hat{A}$ ” FAME axes, not targetome (MH-208)
fame_assay_protein_studies ARM COUNT 96%	by assay class $\hat{A}$: study count DERIVED Defined on: arms with miRTarBase curation / ledger PMIDs $\hat{A}$ ” $\hat{A}$ ” FAME axes, not targetome (MH-208)
fame_assay_reporter_functional_mti_studies ARM COUNT 96%	by assay class $\hat{A}$: functional assays $\hat{A}$: miRNA $\hat{A}$ “target interaction evidence $\hat{A}$: study count DERIVED Defined on: arms with miRTarBase curation / ledger PMIDs $\hat{A}$ ” $\hat{A}$ ” FAME axes, not targetome (MH-208)
fame_assay_reporter_nonfunctional_mti_studies ARM COUNT 96%	by assay class $\hat{A}$: miRNA $\hat{A}$ “target interaction evidence $\hat{A}$: study count DERIVED Defined on: arms with miRTarBase curation / ledger PMIDs $\hat{A}$ ” $\hat{A}$ ” FAME axes, not targetome (MH-208)
fame_assay_reporter_studies ARM COUNT 96%	by assay class $\hat{A}$: study count DERIVED Defined on: arms with miRTarBase curation / ledger PMIDs $\hat{A}$ ” $\hat{A}$ ” FAME axes, not targetome (MH-208)
fame_assay_rna_functional_mti_studies ARM COUNT 96%	by assay class $\hat{A}$: against mRNA $\hat{A}$: functional assays $\hat{A}$: miRNA $\hat{A}$ “target interaction evidence $\hat{A}$: study count DERIVED Defined on: arms with miRTarBase curation / ledger PMIDs $\hat{A}$ ” $\hat{A}$ ” FAME axes, not targetome (MH-208)
fame_assay_rna_functional_mti_weak_studies ARM COUNT 96%	by assay class $\hat{A}$: against mRNA $\hat{A}$: functional assays $\hat{A}$: miRNA $\hat{A}$ “target interaction evidence $\hat{A}$: weak evidence $\hat{A}$: study count DERIVED Defined on: arms with miRTarBase curation / ledger PMIDs $\hat{A}$ ” $\hat{A}$ ” FAME axes, not targetome (MH-208)
fame_assay_rna_nonfunctional_mti_studies ARM COUNT 96%	by assay class $\hat{A}$: against mRNA $\hat{A}$: miRNA $\hat{A}$ “target interaction evidence $\hat{A}$: study count DERIVED Defined on: arms with miRTarBase curation / ledger PMIDs $\hat{A}$ ” $\hat{A}$ ” FAME axes, not targetome (MH-208)
fame_assay_rna_studies ARM COUNT 96%	by assay class $\hat{A}$: against mRNA $\hat{A}$: study count DERIVED Defined on: arms with miRTarBase curation / ledger PMIDs $\hat{A}$ ” $\hat{A}$ ” FAME axes, not targetome (MH-208)
fame_assay_total_studies ARM COUNT 96%	Distinct STUDIES (<code>study_id.unique()</code>) behind this arm's curated edges $\hat{A}$ ” median 57, max 930. See <code>fame_assay_unique_targets</code> : different definitions, one axis (spearman 1.0000). AUTHORED
fame_assay_unique_targets ARM COUNT 96%	Distinct TARGET GENES this arm has in the curated literature (<code>gene.unique()</code>). AUTHORED

fame_led_frac_ltfunc ARM FRACTION 0â€”1 AGG EDGE 96%	fraction DERIVED Defined on: arms with miRTarBase curation / ledger PMIDs â€” â” FAME axes, not targetome (MH-208)
fame_led_n_classes ARM COUNT AGG EDGE 96%	How many distinct evidence CLASSES this arm's curated edges span (1â€”7). AUTHORED
fame_led_n_ltfunc_pmid ARM COUNT AGG EDGE 96%	count â” distinct publications DERIVED Defined on: arms with miRTarBase curation / ledger PMIDs â€” â” FAME axes, not targetome (MH-208)
fame_led_n_weak ARM COUNT AGG EDGE 94%	count â” weak evidence DERIVED Defined on: arms with miRTarBase curation / ledger PMIDs â€” â” FAME axes, not targetome (MH-208)
fame_n_genes_curated ARM COUNT AGG EDGE 96%	count â” genes DERIVED Defined on: arms with miRTarBase curation / ledger PMIDs â€” â” FAME axes, not targetome (MH-208)
fame_npmid ARM COUNT AGG EDGE 96%	Distinct PMIDs citing this arm. AUTHORED

AGO_ â” 4
supporting â€” read these when the core raises a question

4 columns â” coverage **28â€”60%** (median 28%)
1x fraction 0â€”1 â” 1x constant â” 1x categorical â” 1x real â‰¥ 0

ago_dom ARM FRACTION 0â€”1 28%	Fraction of the arm's Manakov AGO-CLIP reads assigned to it rather than its sibling arm â€” the loading dominance. Median 0.658 on the 27.7% of arms with CLIP coverage. AUTHORED
ago_dom_src ARM CONSTANT 28%	Provenance of ago_dom â€” which CLIP dataset the AGO-loading dominance came from. AUTHORED
ago_guide_class ARM CATEGORICAL 28%	AGO/RISC loading gate for the arm. AUTHORED
ago_reads ARM REAL â‰¥ 0 60%	Raw AGO-CLIP read count (median 6, max 210,722). AUTHORED

HLY_ â” 7
supporting â€” read these when the core raises a question

HEALTHY baseline levels for the arm â€” GTEx breast median/percentile and TCGA-NAT median.
7 columns â” coverage **17â€”91%** (median 91%) â” **3** share the block description
2x real â‰¥ 0 â” 1x categorical â” 1x boolean â” 1x percent â” 1x count â” 1x fraction 0â€”1

hly_baseline_src ARM CATEGORICAL 91%	source DERIVED Defined on: arms with a GTEx healthy anchor â€” â” MEASURED leg only; floor0 is NEVER emitted as a value
hly_from_seedmate ARM BOOLEAN 91%	Was this arm's healthy baseline borrowed from a SEED-MATE rather than measured directly (49 arms)? AUTHORED
hly_gtex_median ARM REAL â‰¥ 0 17%	GTEx â” median DERIVED Defined on: arms with a GTEx healthy anchor â€” â” MEASURED leg only; floor0 is NEVER emitted as a value

hly_gtex_pct ARM PERCENT 17%	GTEX % percent DERIVED Defined on: arms with a GTEX healthy anchor as MEASURED leg only; floor0 is NEVER emitted as a value
hly_nat_detect_n ARM COUNT 91%	In how many of the ~104 matched NAT samples this arm is detected above the floor. AUTHORED
hly_nat_median ARM REAL % 0 74%	Median NAT abundance (fill 74.0%). AUTHORED
hly_tumor_prev ARM FRACTION 01 91%	Prevalence of the arm above the detection floor in tumour (median 0.01). AUTHORED

DOSE_ 3

supporting read these when the core raises a question

Dose (abundance) of the regulator and how it shifts across states, plus host-gene co-transcription fractions where the arm is intragenic.

3 columns coverage 12%12% (median 12%) 2 share the block description

2x real % 0 1x constant

dose_comp_retention ARM REAL % 0 12%	retention DERIVED Defined on: edges with a matched NAT leg (n~104 paired participants)
dose_confounded ARM CONSTANT 12%	Is this arm's NAT's tumour dose a composition or proliferation artifact (either gated retention < 0.4)? AUTHORED
dose_prolif_retention ARM REAL % 0 12%	retention DERIVED Defined on: edges with a matched NAT leg (n~104 paired participants)

ABUND_ 5

supporting read these when the core raises a question

The arm's own abundance profile across the cohort as median, SD, IQR, detection fraction.

5 columns coverage 88%91% (median 91%) 4 share the block description

3x real % 0 1x fraction 01 1x categorical

abund_frac_detected ARM FRACTION 01 91%	fraction DERIVED Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)
abund_iqr ARM REAL % 0 91%	interquartile range DERIVED Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)
abund_med ARM REAL % 0 91%	median DERIVED Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)
abund_sd ARM REAL % 0 88%	standard deviation DERIVED Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)
abund_sd_tertile ARM CATEGORICAL 88%	Dispersion tertile as 718 / 716 / 718 by construction, so it carries no information beyond abund_sd's rank. AUTHORED

ATTR_ 14

reference “ context, provenance and rarely-quoted detail

Attribution rollup over the arm's edges “ how much of the gene's coupling this arm is credited with, and over how many reliable edges.

14 columns coverage 6%–29% (median 15%) 9 share the block description

11× fraction 0–1 2× count 1× signed –1|1

attr_max_budget_share ARMFRACTION 0–115%	maximum pressure budget share DERIVED Defined on: arms with a Shapley/attribution decomposition over their edges (728)
attr_max_credit_share ARMFRACTION 0–16%	maximum share DERIVED Defined on: arms with a Shapley/attribution decomposition over their edges (728)
attr_max_joint_share ARMFRACTION 0–115%	Largest JOINT attribution share among this arm's edges. AUTHORED
attr_max_nested_share ARMFRACTION 0–115%	Largest NESTED attribution share. See attr_max_joint_share : different values (agree on 35.4%), one axis (spearman +0.962). AUTHORED Defined on: arms with a Shapley/attribution decomposition over their edges (728)
attr_max_within_share ARMFRACTION 0–115%	Largest WITHIN-cell attribution share for this arm. AUTHORED
attr_med_budget_share ARMFRACTION 0–115%	median pressure budget share DERIVED Defined on: arms with a Shapley/attribution decomposition over their edges (728)
attr_med_credit_share ARMFRACTION 0–16%	median share DERIVED Defined on: arms with a Shapley/attribution decomposition over their edges (728)
attr_med_dose_share ARMFRACTION 0–129%	Median dose share of this arm within its family cells. AUTHORED
attr_med_joint_share ARMFRACTION 0–115%	median share DERIVED Defined on: arms with a Shapley/attribution decomposition over their edges (728)
attr_med_nested_share ARMFRACTION 0–115%	median share DERIVED Defined on: arms with a Shapley/attribution decomposition over their edges (728)
attr_med_phi_joint ARMSIGNED –1 115%	median DERIVED Defined on: arms with a Shapley/attribution decomposition over their edges (728)
attr_med_within_share ARMFRACTION 0–115%	median share DERIVED Defined on: arms with a Shapley/attribution decomposition over their edges (728)
attr_n_edges ARMCOUNT22%	count DERIVED Defined on: arms with a Shapley/attribution decomposition over their edges (728)
attr_n_reliable ARMCOUNT15%	Reliable edges behind this arm's attr_* shares (median 2). AUTHORED

CTX_ 2

reference " context, provenance and rarely-quoted detail

Gene-design context from the decoy/atlas layer (`learned/analyses/gene_atlas.py` via `card_context`): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.

2 columns coverage 29% (median 29%) 2 share the block description

1x boolean 1x real 0

ctx_arm_abundant ARM BOOLEAN 29%	per arm DERIVED Defined on: arms present on the EDGE card (729 of 2,450) " lifted arm-rung columns, names preserved
---	--

ctx_arm_dose ARM REAL 0 29%	per arm dose DERIVED Defined on: arms present on the EDGE card (729 of 2,450) " lifted arm-rung columns, names preserved
--	---

GRANK_ 3

reference " context, provenance and rarely-quoted detail

The arm's GLOBAL (cohort-wide) dose rank in one state. This is what the dominant-regulator and handoff columns are computed from.

3 columns coverage 15% (median 20%) 3 share the block description

3x count

grank_HLY ARM COUNT 15%	in the GTEx-healthy state DERIVED Defined on: arms with a GTEx-HEALTHY leg (the thinnest state " many arms are multi-mapping-collapsed, MH-166)
grank_NAT ARM COUNT 20%	in matched normal (NAT) DERIVED Defined on: edges with a matched NAT leg (n~104 paired participants)
grank_TUM ARM COUNT 20%	in tumour DERIVED Defined on: edges present in the progression panel (arm expressed + state measured)

REAL_ 11

reference " context, provenance and rarely-quoted detail

Realization rollout for the ARM " how many of its edges are scored, and its median and best realized coupling.

11 columns coverage 9% (median 24%) 8 share the block description

7x count 2x signed 1 2x fraction 01

real_best_coupling ARM SIGNED 11 24%	Most negative realized coupling among this arm's scored genes (median 0.08, min 0.56). AUTHORED
real_frac_c10 ARM FRACTION 01 9%	Fraction of this arm's scored genes coupling below 0.10. AUTHORED
real_frac_sig ARM FRACTION 01 9%	fraction DERIVED Defined on: arms with 1 calibrated-coupling-scored edge (593) " the realization LADDER; rates are gated at 5
real_med_coupling ARM SIGNED 11 24%	median DERIVED Defined on: arms with 1 calibrated-coupling-scored edge (593) " the realization LADDER; rates are gated at 5

real_n_c05 ARM COUNT 24%	count $\hat{A}$ at the $\hat{a}^{*0.05}$ threshold DERIVED <i>Defined on: arms with $\hat{a}^{*0.05}$ calibrated-coupling-scored edge (593) $\hat{a}^{*0.05}$ the realization LADDER; rates are gated at $\hat{a}^{*0.05}$</i>
real_n_c10 ARM COUNT 24%	count $\hat{A}$ at the $\hat{a}^{*0.10}$ threshold DERIVED <i>Defined on: arms with $\hat{a}^{*0.10}$ calibrated-coupling-scored edge (593) $\hat{a}^{*0.10}$ the realization LADDER; rates are gated at $\hat{a}^{*0.10}$</i>
real_n_c15 ARM COUNT 24%	count $\hat{A}$ at the $\hat{a}^{*0.15}$ threshold DERIVED <i>Defined on: arms with $\hat{a}^{*0.15}$ calibrated-coupling-scored edge (593) $\hat{a}^{*0.15}$ the realization LADDER; rates are gated at $\hat{a}^{*0.15}$</i>
real_n_c20 ARM COUNT 24%	count $\hat{A}$ at the $\hat{a}^{*0.20}$ threshold DERIVED <i>Defined on: arms with $\hat{a}^{*0.20}$ calibrated-coupling-scored edge (593) $\hat{a}^{*0.20}$ the realization LADDER; rates are gated at $\hat{a}^{*0.20}$</i>
real_n_c30 ARM COUNT 24%	count $\hat{A}$ at the $\hat{a}^{*0.30}$ threshold DERIVED <i>Defined on: arms with $\hat{a}^{*0.30}$ calibrated-coupling-scored edge (593) $\hat{a}^{*0.30}$ the realization LADDER; rates are gated at $\hat{a}^{*0.30}$</i>
real_n_scored ARM COUNT 24%	Genes scored for this arm's realized coupling (median 3). $\hat{a}^{*0.30}$ Its <code>n_c05</code> $\hat{a}^{*0.30}$ ladder is properly nested $\hat{a}^{*0.30}$ 0 violations (verified). AUTHORED <i>Defined on: arms with $\hat{a}^{*0.30}$ calibrated-coupling-scored edge (593) $\hat{a}^{*0.30}$ the realization LADDER; rates are gated at $\hat{a}^{*0.30}$</i>
real_n_sig ARM COUNT 24%	count DERIVED <i>Defined on: arms with $\hat{a}^{*0.30}$ calibrated-coupling-scored edge (593) $\hat{a}^{*0.30}$ the realization LADDER; rates are gated at $\hat{a}^{*0.30}$</i>

TS_ $\hat{A}$ 10
reference $\hat{a}^{*0.30}$ context, provenance and rarely-quoted detail

TargetScan context++ scores for the arm's sites $\hat{a}^{*0.30}$ number of sites, mean and best context score. More negative context++ = stronger predicted repression.

10 columns $\hat{A}$ coverage **13%** $\hat{a}^{*0.30}$ 13% (median 13%) $\hat{A}$ **8** share the block description

5x count $\hat{A}$ 2x real, signed $\hat{A}$ 1x signed $\hat{a}^{*0.30}$ 1 $\hat{A}$ 1x fraction 0 $\hat{a}^{*0.30}$ 1 $\hat{A}$ 1x real $\hat{a}^{*0.30}$ 0

ts_ctx_best ARM REAL, SIGNED 13%	Best (most negative) TargetScan context++ score for this arm $\hat{a}^{*0.30}$ range [$\hat{a}^{*2.02}$, $\hat{a}^{*0.34}$], all negative by construction (context++ scores repression as a negative number). AUTHORED
ts_ctx_mean ARM SIGNED $\hat{A}^{*0.30}$1 13%	mean DERIVED <i>Defined on: arms in TargetScan's human default predictions ($\hat{a}^{*0.30}$ only 321)</i>
ts_frac_8mer ARM FRACTION 0$\hat{a}^{*0.30}$1 13%	fraction $\hat{A}$ 8mer sites DERIVED <i>Defined on: arms in TargetScan's human default predictions ($\hat{a}^{*0.30}$ only 321)</i>
ts_kd_med ARM REAL, SIGNED 13%	median DERIVED <i>Defined on: arms in TargetScan's human default predictions ($\hat{a}^{*0.30}$ only 321)</i>
ts_n_7mer_A1 ARM COUNT 13%	count $\hat{A}$ 7mer sites DERIVED <i>Defined on: arms in TargetScan's human default predictions ($\hat{a}^{*0.30}$ only 321)</i>
ts_n_7mer_m8 ARM COUNT 13%	count $\hat{A}$ 7mer sites DERIVED <i>Defined on: arms in TargetScan's human default predictions ($\hat{a}^{*0.30}$ only 321)</i>
ts_n_8mer ARM COUNT 13%	count $\hat{A}$ 8mer sites DERIVED <i>Defined on: arms in TargetScan's human default predictions ($\hat{a}^{*0.30}$ only 321)</i>

ts_n_genes ARM COUNT AGG EDGE 13%	Genes with a TargetScan context++ site “ median 497, fill 12.9%, the SPARSEST universe on the card . Correlates +0.726 with the MANE seed scan and only +0.231 with scanMiR-any. AUTHORED
ts_n_sites ARM COUNT AGG EDGE 13%	count DERIVED Defined on: arms in TargetScan's human default predictions (â only 321)
ts_sites_per_gene ARM REAL “ 0 13%	per gene DERIVED Defined on: arms in TargetScan's human default predictions (â only 321)

ISOC_ “ 5

reference “ context, provenance and rarely-quoted detail

isomiR COMPOSITION “ what fraction of the arm's reads are canonical, orphan, or donated from a neighbouring arm.

5 columns “ coverage **91%“91%** (median 91%) “ 4 share the block description

3× fraction 0“1 “ 1× real “ 0 “ 1× count

isoc_canonical_frac ARM FRACTION 0“1 91%	canonical sites “ fraction DERIVED Defined on: arms in the isomiR seed-COMPOSITION table “ where the reads go; â isoc_orphan_frac flags MH-153 mass loss
isoc_donated_frac ARM FRACTION 0“1 91%	fraction DERIVED Defined on: arms in the isomiR seed-COMPOSITION table “ where the reads go; â isoc_orphan_frac flags MH-153 mass loss
isoc_med_reads ARM REAL “ 0 91%	Median isomiR reads for this arm “ median 1.00, max 528,927 . AUTHORED
isoc_n_dest_families ARM COUNT 91%	count DERIVED Defined on: arms in the isomiR seed-COMPOSITION table “ where the reads go; â isoc_orphan_frac flags MH-153 mass loss
isoc_orphan_frac ARM FRACTION 0“1 91%	fraction DERIVED Defined on: arms in the isomiR seed-COMPOSITION table “ where the reads go; â isoc_orphan_frac flags MH-153 mass loss

SDSZ_ “ 4

reference “ context, provenance and rarely-quoted detail

MANE 3'UTR seed scan.

4 columns “ coverage **97%“97%** (median 97%) “ 2 share the block description

2× count “ 1× real “ 0 “ 1× fraction 0“1

sdsz_N_7mer_plus ARM COUNT AGG EDGE 97%	count “ 7mer sites DERIVED Defined on: arms in the MANE 3'UTR seed scan “ â a FOURTH targetome universe, never blend (2,370)
sdsz_N_8mer ARM COUNT AGG EDGE 97%	count “ 8mer sites DERIVED Defined on: arms in the MANE 3'UTR seed scan “ â a FOURTH targetome universe, never blend (2,370)
sdsz_N_eff ARM REAL “ 0 AGG EDGE 97%	Effective seed-site count over the scanned universe (median 2,094, fill 96.7% “ the best-covered targetome axis). AUTHORED

sdsz_rarity

The arm's seed RARITY, min-max normalised to 0-1 (median 0.11).

AUTHORED

ARM

FRACTION 0-1

97%

CHIM_ 5

reference context, provenance and rarely-quoted detail

Chimeric-evidence ROLLUP for the arm (aggregated over its edges).

5 columns coverage 21-63% (median 60%) 3 share the block description
4x count 1x real 0

chim_manakov_n_genes

count genes DERIVED

ARM COUNT AGG EDGE 60%

Defined on: arms with chimeric ligation evidence (1,538) ~90% Manakov/HEK293T, per-source, NEVER pooled

chim_manakov_weight

Manakov chimeric support summed over this arm's edges (median 6, max 210,722).

AUTHORED

ARM REAL 0 AGG EDGE 60%

chim_n_sources

count DERIVED

ARM COUNT AGG EDGE 63%

Defined on: arms with chimeric ligation evidence (1,538) ~90% Manakov/HEK293T, per-source, NEVER pooled

chim_tarbase_n_genes

count genes DERIVED

ARM COUNT AGG EDGE 21%

Defined on: arms with chimeric ligation evidence (1,538) ~90% Manakov/HEK293T, per-source, NEVER pooled

chim_tarbase_weight

TarBase chimeric support for the arm (median 8, max 10,255), defined on 20.9% of arms a third of the Manakov coverage.

AUTHORED

ARM COUNT AGG EDGE 21%

Defined on: arms with chimeric ligation evidence (1,538) ~90% Manakov/HEK293T, per-source, NEVER pooled

FST_ 16

reference context, provenance and rarely-quoted detail

THE ARM'S SHARE OF ITS OWN SEED FAMILY, ACROSS STATES gene-free (card_ladders.family_state_shift , 2026-08-19).

16 columns coverage 3-100% (median 20%) 7 share the block description
6x count 5x boolean 3x fraction 0-1 2x signed -1

fst_d_share_HLY_TUM

As fst_d_share_NAT_TUM but GTEx-healthy tumour, same common-support correction. n=68 (HLY is the sparsest leg at 17.0%). Post-fix mean 0.0000, 47.1% positive, p=0.93.

AUTHORED

ARM SIGNED -1 3%

fst_d_share_NAT_TUM

Change in the arm's share of its seed family's linear dose, matched NAT tumour, re-normalised over the members measured in BOTH states.

AUTHORED

ARM SIGNED -1 6%

fst_dominance_switch

Did the family's DOMINANT arm change between healthy and tumour (103 true / 74 false)?

AUTHORED

ARM BOOLEAN 7%

fst_hly_family_measured

Do ALL members of this arm's family have a measured GTEx level.

AUTHORED

ARM BOOLEAN 100%

fst_hly_measured

in the GTEx-healthy state

DERIVED

ARM BOOLEAN 100%

fst_is_dominant_HLY ARM BOOLEAN 17%	the dominant one $\hat{\cdot}$ in the GTEx-healthy state DERIVED
fst_is_dominant_TUM ARM BOOLEAN 20%	the dominant one $\hat{\cdot}$ in tumour DERIVED
fst_n_common_HLY_TUM ARM COUNT 9%	Members measured in BOTH GTEx-healthy and tumour $\hat{\cdot}$ the denominator behind <code>fst_d_share_HLY_TUM</code> . See <code>fst_n_common_NAT_TUM</code> . AUTHORED
fst_n_common_NAT_TUM ARM COUNT 20%	How many of the family's arms are measured in BOTH NAT and tumour $\hat{\cdot}$ the DENOMINATOR behind <code>fst_d_share_NAT_TUM</code> , shipped because the column was wrong for want of it. AUTHORED
fst_n_members ARM COUNT 100%	count $\hat{\cdot}$ family members DERIVED
fst_rank_HLY ARM COUNT 17%	rank $\hat{\cdot}$ in the GTEx-healthy state DERIVED
fst_rank_NAT ARM COUNT 74%	rank $\hat{\cdot}$ in matched normal (NAT) DERIVED
fst_rank_TUM ARM COUNT 20%	rank $\hat{\cdot}$ in tumour DERIVED
fst_share_HLY ARM FRACTION $\hat{\cdot}$ 1 17%	Share of family dose in GTEx-healthy, from <code>hly_gtex_median</code> (fill 17.0% $\hat{\cdot}$ the sparsest leg). Same singleton degeneracy; and cross-platform, so contrast only. AUTHORED
fst_share_NAT ARM FRACTION $\hat{\cdot}$ 1 74%	Share of family dose in matched NAT, from <code>hly_nat_median</code> (fill 74.0% $\hat{\cdot}$ the best-covered leg, 3.7 $\hat{\cdot}$ TUM). Same singleton degeneracy: median 1.000, mask on <code>fst_n_members > 1</code> . AUTHORED
fst_share_TUM ARM FRACTION $\hat{\cdot}$ 1 20%	The arm's fraction of its seed family's total linear dose in tumour, from <code>aim_med_rpm</code> (fill 19.8%). AUTHORED

SEQ_ $\hat{\cdot}$ 4

reference $\hat{\cdot}$ context, provenance and rarely-quoted detail

4 columns $\hat{\cdot}$ coverage 30%$\hat{\cdot}$30% (median 30%) 2$\times$ count $\hat{\cdot}$ 2$\times$ real $\hat{\cdot}$ 0	
seq_n_genes_any ARM COUNT AGG EDGE 30%	Genes with ANY scanMiR K_D site $\hat{\cdot}$ AUTHORED
seq_n_genes_strong ARM COUNT AGG EDGE 30%	Genes with a STRONG scanMiR K_D site $\hat{\cdot}$ median 3,632, spearman +0.822 with <code>sdsz_N_eff</code> and +0.797 with <code>site_n_genes_canonical</code> , but only +0.566 with <code>ts_n_genes</code> . AUTHORED
seq_promisc ARM REAL $\hat{\cdot}$ 0 30%	log-scale promiscuity over STRONG scanMiR sites (4.45 $\hat{\cdot}$ 8.94). AUTHORED
seq_promisc_any ARM REAL $\hat{\cdot}$ 0 30%	Promiscuity over ANY scanMiR site (6.46 $\hat{\cdot}$ 9.45) $\hat{\cdot}$ the permissive cut. AUTHORED

SUB_ 6

reference " context, provenance and rarely-quoted detail

PAM50 subtype block aggregated to the ARM. See `esub_` for the per-edge form.

6 columns coverage 30%30% (median 30%) 4 share the block description

2x count 2x signed 11 1x categorical 1x fraction 01

sub_best_frac ARMFRACTION 0130%	the best fraction DERIVED Defined on: arms with a per-PAM50 edge-heterogeneity fit (728) " DISTRIBUTIONAL (MH-165), not a validated label
sub_best_mode ARMCATEGORICAL 30%	the best DERIVED Defined on: arms with a per-PAM50 edge-heterogeneity fit (728) " DISTRIBUTIONAL (MH-165), not a validated label
sub_med_rho_best ARMSIGNED 1130%	Median coupling in the arm's BEST subtype. AUTHORED
sub_med_rho_pool ARMSIGNED 1130%	Median pooled-subtype coupling (median 0.0064) " the unselected reference for <code>sub_med_rho_best</code>. AUTHORED Defined on: arms with a per-PAM50 edge-heterogeneity fit (728) " DISTRIBUTIONAL (MH-165), not a validated label
sub_n_edges ARMCOUNT 30%	count DERIVED Defined on: arms with a per-PAM50 edge-heterogeneity fit (728) " DISTRIBUTIONAL (MH-165), not a validated label
sub_n_q05 ARMCOUNT 30%	count DERIVED Defined on: arms with a per-PAM50 edge-heterogeneity fit (728) " DISTRIBUTIONAL (MH-165), not a validated label

COV_ 22

reference " context, provenance and rarely-quoted detail

Coverage flags " was this row ever SCANNED by the named source.

22 columns coverage 100%100% (median 100%) 2 share the block description

22x boolean

cov_ago_dom ARMBOOLEAN 100%	Was this arm ever SCANNED for AGO-loading dominance from Manakov CLIP? True on 678 of 2,450 arms. AUTHORED
cov_attr ARMBOOLEAN 100%	Was this arm ever SCANNED for the Shapley/NNLS attribution block? True on 546 of 2,450 arms. AUTHORED
cov_beta ARMBOOLEAN 100%	$\hat{\rho}^2$ DERIVED
cov_cnv ARMBOOLEAN 100%	Was this arm ever SCANNED for the arm-locus copy-number block? True on 856 of 2,450 arms. AUTHORED
cov_comp ARMBOOLEAN 100%	Was this arm ever SCANNED for the compartment-loading (<code>cload_</code>) block? True on 205 of 2,450 arms. AUTHORED
cov_cptac ARMBOOLEAN 100%	CPTAC DERIVED Defined on: arms measurable in CPTAC (2,095)

cov_curated ARM BOOLEAN 100%	Was this arm ever SCANNED for the curated-literature blocks (fame_ , cur_)? True on 875 of 2,450 arms. AUTHORED
cov_expr ARM BOOLEAN 100%	Was this arm ever SCANNED for the expression / abundance blocks? True on 2,231 of 2,450 arms. AUTHORED
cov_famrole ARM BOOLEAN 100%	Was this arm ever SCANNED for the family-role block? True on 2,231 of 2,450 arms. AUTHORED
cov_field ARM BOOLEAN 100%	Was this arm ever SCANNED for the focal-field correlation block? True on 571 of 2,450 arms. AUTHORED
cov_focal ARM BOOLEAN 100%	Was this arm ever SCANNED for the focal-locus CNV block? True on 733 of 2,450 arms. AUTHORED
cov_fst ARM BOOLEAN 100%	Was the family-share block computed for this arm at all â€” i.e. did it have the abundance input. AUTHORED
cov_gctx ARM BOOLEAN 100%	Was this arm ever SCANNED for the genomic-context block? True on 714 of 2,450 arms. AUTHORED
cov_isocomp ARM BOOLEAN 100%	Was this arm ever SCANNED for the isomiR-composition block? True on 2,222 of 2,450 arms. AUTHORED
cov_isomir ARM BOOLEAN 100%	Was this arm ever SCANNED for the isomiR seed-shift block? True on 687 of 2,450 arms. AUTHORED
cov_meth ARM BOOLEAN 100%	Was this arm ever SCANNED for the locus-methylation (bc_meth_) block? True on 343 of 2,450 arms. AUTHORED
cov_real ARM BOOLEAN 100%	Was this arm ever SCANNED for the realized-coupling (real_) block? True on 590 of 2,450 arms. AUTHORED
cov_seed_scan ARM BOOLEAN 100%	Was this arm ever SCANNED for the MANE 3'UTR seed scan (site_)? AUTHORED
cov_seq_scan ARM BOOLEAN 100%	Was this arm ever SCANNED for the scanMiR K_D scan (seq_)? AUTHORED
cov_site_types ARM BOOLEAN 100%	Was this arm ever SCANNED for the site-type breakdown? AUTHORED
cov_subtype ARM BOOLEAN 100%	Was this arm ever SCANNED for the PAM50 subtype block (sub_)? AUTHORED
cov_targetscan ARM BOOLEAN 100%	Was this arm ever SCANNED for the TargetScan context++ block (ts_)? AUTHORED

CUR_ 2

reference â€” context, provenance and rarely-quoted detail

2 columns

2x count

coverage 36%â€”36% (median 36%)

cur_he_degree How many HE genes this arm regulates in the curated design – the arm's curation degree (median 3, max 125). AUTHORED

ARM COUNT AGG EDGE 36%

Defined on: arms with miRTarBase curation / ledger PMIDs – FAME axes, not targetome (MH-208)

cur_he_degree_expr As **cur_he_degree** but counting only targets above the expression floor. AUTHORED

ARM COUNT AGG EDGE 36%

ISO_ 5

reference – context, provenance and rarely-quoted detail

isomiR profile for the arm – 5' shift fraction and seed-shift occurrence. Drives the isomiR-corrected X_fam, which is BUILT but default-OFF (coupling wash, bidirectional).

5 columns – coverage **28%–28%** (median 28%) – 4 share the block description
2× fraction 0–1 – 1× count – 1× real – 0 – 1× real, signed

iso_n_samples count DERIVED

ARM COUNT 28%

Defined on: arms with a 5'-isomiR seed-shift measurement (687) – gate on iso_n_samples, the median is n-DEPENDENT

iso_rpm_summed – A SUM ACROSS SAMPLES, not a per-sample RPM – the name misleads. Max **270,988,199**, which is impossible for a per-million unit; divided by **iso_n_samples** the scale is sane (median 57, max 251,381). AUTHORED

ARM REAL – 0 28%

iso_seed_shift_frac shift – fraction DERIVED

ARM FRACTION 0–1 28%

Defined on: arms with a 5'-isomiR seed-shift measurement (687) – gate on iso_n_samples, the median is n-DEPENDENT

iso_seed_shift_sd shift – standard deviation DERIVED

ARM FRACTION 0–1 28%

Defined on: arms with a 5'-isomiR seed-shift measurement (687) – gate on iso_n_samples, the median is n-DEPENDENT

iso_top_shift_nt the top-ranked – shift DERIVED

ARM REAL, SIGNED 28%

Defined on: arms with a 5'-isomiR seed-shift measurement (687) – gate on iso_n_samples, the median is n-DEPENDENT

FIELD_ 5

reference – context, provenance and rarely-quoted detail

Within-patient NAT field-effect fit – is the regulator already established in the patient's adjacent normal before the tumour. Defined on the paired subset.

5 columns – coverage **23%–23%** (median 23%) – 3 share the block description
4× signed –1–1 – 1× count

field_excess_over_perm – NOT A RETENTION RATIO – a FIFTH sense of the word in this codebase. Measured: it equals **field_r_own – field_r_perm** (exact within rounding on 100% of 571 arms, max residual 0.001), i.e. AUTHORED

ARM SIGNED –1–1 23%

field_n count DERIVED

ARM COUNT 23%

Defined on: arms with a within-patient NAT field-effect fit (571)

field_r_adj correlation – composition-adjusted DERIVED

ARM SIGNED –1–1 23%

Defined on: arms with a within-patient NAT field-effect fit (571)

field_r_own correlation DERIVED

ARM SIGNED –1–1 23%

Defined on: arms with a within-patient NAT field-effect fit (571)

field_r_perm

The permutation null for field_r_own " median 0.000, range [0.11, +0.09].

AUTHORED

ARM

SIGNED -11

23%

FAMROLE_ 6

reference " context, provenance and rarely-quoted detail

The arm's ROLE inside its seed family " member count, its share of family abundance, and its rank among members.

6 columns " coverage 88%100% (median 91%) " 5 share the block description

3x count " 1x fraction 01 " 1x categorical " 1x boolean

famrole_abund_rank ARM COUNT 91%	unweighted abundance-sum reference " rank DERIVED Defined on: arms with a seed family " the arm's ROLE inside it; " DEGENERATE at famrole_n_members==1
famrole_abund_share ARM FRACTION 01 91%	unweighted abundance-sum reference " share DERIVED Defined on: arms with a seed family " the arm's ROLE inside it; " DEGENERATE at famrole_n_members==1
famrole_class ARM CATEGORICAL 100%	The arm's role in its seed family " sole on 1,639 of 2,450 (66.9%), dominant 282, co and the rest below. AUTHORED
famrole_is_dominant ARM BOOLEAN 91%	the dominant one DERIVED Defined on: arms with a seed family " the arm's ROLE inside it; " DEGENERATE at famrole_n_members==1
famrole_n_members ARM COUNT 100%	count " family members DERIVED Defined on: arms with a seed family " the arm's ROLE inside it; " DEGENERATE at famrole_n_members==1
famrole_var_rank ARM COUNT 88%	rank DERIVED Defined on: arms with a seed family " the arm's ROLE inside it; " DEGENERATE at famrole_n_members==1

BC_ 11

reference " context, provenance and rarely-quoted detail

Breast-cancer locus annotation for the arm " CpG probes, CGI overlap and related epigenetic context at the precursor locus.

11 columns " coverage 14%36% (median 14%) " 9 share the block description

4x fraction 01 " 4x count " 1x boolean " 1x signed -11 " 1x categorical

bc_fam_context_homogeneous ARM BOOLEAN 36%	per seed family DERIVED Defined on: " BROADCAST from the seed FAMILY (genomic_context.family_context)
bc_fam_dose_coding_host_fraction ARM FRACTION 01 36%	per seed family " dose " fraction DERIVED Defined on: " BROADCAST from the seed FAMILY (genomic_context.family_context)
bc_fam_dose_cotranscribed_fraction ARM FRACTION 01 36%	per seed family " dose " fraction DERIVED Defined on: " BROADCAST from the seed FAMILY (genomic_context.family_context)
bc_fam_n_distinct_coding_hosts ARM COUNT 36%	per seed family " count DERIVED Defined on: " BROADCAST from the seed FAMILY (genomic_context.family_context)

bc_fam_n_members ARMCOUNT36%	per seed family Â· count Â· family members DERIVED Defined on: âš BROADCAST from the seed FAMILY (genomic_context.family_context)
bc_meth_dbeta ARMSIGNED -1114%	Tumour-minus-normal methylation beta at this arm's locus (median âˆ’0.03, range âˆ’0.32â€¦+0.23). AUTHORED
bc_meth_direction ARMCATEGORICAL14%	Tumour-vs-normal methylation direction at this arm's locus. AUTHORED
bc_meth_n_cgi ARMCOUNT14%	count DERIVED Defined on: âš BROADCAST from the pri-miRNA HAIRPIN â€” 5p and 3p share one promoter and one Î”Î² (354 arms)
bc_meth_n_probes ARMCOUNT14%	count DERIVED Defined on: âš BROADCAST from the pri-miRNA HAIRPIN â€” 5p and 3p share one promoter and one Î”Î² (354 arms)
bc_meth_normal_beta ARMFRACTION 0114%	Î² DERIVED Defined on: âš BROADCAST from the pri-miRNA HAIRPIN â€” 5p and 3p share one promoter and one Î”Î² (354 arms)
bc_meth_tumour_beta ARMFRACTION 0114%	Î² DERIVED Defined on: âš BROADCAST from the pri-miRNA HAIRPIN â€” 5p and 3p share one promoter and one Î”Î² (354 arms)

GCTX_ Â· 16
reference â€” context, provenance and rarely-quoted detail

Genomic context of the arm's precursor locus (<code>genomic_context.classify_he_arms</code>): host gene, sense/antisense orientation, intron-vs-exon, promoter class. Strand-aware. 16 columns Â· coverage 24%â€”29% (median 29%) Â· 6 share the block description 6× categorical Â· 4× boolean Â· 3× identifier Â· 1× constant Â· 1× count Â· 1× real â‰¥ 0	
gctx_clustered ARMBOOLEAN29%	Does this arm's precursor sit in a genomic CLUSTER with other miRNAs (310 true / 404 false)? AUTHORED
gctx_context_mixed ARMBOOLEAN29%	Do this arm's multiple loci DISAGREE about their genomic context â€” 20 arms of 714? AUTHORED
gctx_coord_source ARMCONSTANT29%	Provenance of the genomic-context coordinates. AUTHORED
gctx_host ARMIDENTIFIER27%	The HOST GENE the precursor sits inside (356 distinct; <code>ENSG00000269842</code> , <code>MEG9</code> , <code>MEG3</code> most common). AUTHORED
gctx_host_region ARMIDENTIFIER27%	WHERE in the host the precursor sits (74 distinct; <code>intron 2</code> and <code>intron 1</code> most common, 61 arms each). An intronic miRNA is spliced from its host's pre-mRNA; an exonic one competes with it. AUTHORED Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€” 714)
gctx_host_type ARMCATEGORICAL27%	The host's BIOTYPE â€” <code>protein_coding</code> 364 Â· <code>lncRNA</code> 303, plus 5 rarer levels. AUTHORED
gctx_in_exon ARMBOOLEAN29%	Is the precursor inside a host EXON (218 true / 496 false) rather than an intron? AUTHORED

gctx_mir_class ARM CATEGORICAL29%	class label DERIVED Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€” 714)
gctx_mirgenedb ARM BOOLEAN29%	Is this arm in MirGeneDB â€” the curated high-confidence miRNA set (526 true / 188 false)? AUTHORED
gctx_n_loci ARM COUNT29%	count DERIVED Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€” 714)
gctx_promoter_class ARM CATEGORICAL29%	class label DERIVED Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€” 714)
gctx_promoter_coord_class ARM CATEGORICAL29%	class label DERIVED Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€” 714)
gctx_promoter_gene ARM IDENTIFIER24%	per gene DERIVED Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€” 714)
gctx_promoter_gene_type ARM CATEGORICAL24%	per gene DERIVED Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€” 714)
gctx_promoter_host_tss_kb ARM REAL 024%	Distance in kb from the precursor to its host's TSS (median 0.5 kb, max 874). AUTHORED
gctx_strand_rel ARM CATEGORICAL27%	Is the precursor SENSE (633 arms) or ANTISENSE (40) to its host? AUTHORED

SURV_ 8

reference â€” context, provenance and rarely-quoted detail

TCGA outcome panel legs. 8 columns â€” coverage 6%â€”6% (median 6%) â€” 5 share the block description 4x p-value â€” 3x real 0 â€” 1x constant	
surv_dfi_hr_adj ARM REAL 06%	Adjusted DFI hazard ratio per arm â€” range [0.84, 1.16], median 1.00. The whole distribution sits within Â±16% of no effect. See surv_dfi_q_adj. DERIVED Defined on: arms in the TCGA outcome panel (145) â€” TCGA legs only; the GSE leg is bare-stem matched
surv_dfi_hr_base ARM REAL 06%	Unadjusted DFI hazard ratio for the arm (0.86 â€” 1.16, median 1.00). AUTHORED
surv_dfi_p_adj ARM P-VALUE6%	p-value â€” composition-adjusted DERIVED Defined on: arms in the TCGA outcome panel (145) â€” TCGA legs only; the GSE leg is bare-stem matched
surv_dfi_p_base ARM P-VALUE6%	p-value DERIVED Defined on: arms in the TCGA outcome panel (145) â€” TCGA legs only; the GSE leg is bare-stem matched
surv_dfi_q_adj ARM CONSTANT6%	BH q for the arm's DFI hazard ratio. AUTHORED
surv_os_hr_adj ARM REAL 06%	composition-adjusted DERIVED Defined on: arms in the TCGA outcome panel (145) â€” TCGA legs only; the GSE leg is bare-stem matched

surv_os_p_adj ARM P-VALUE 6%	p-value $\hat{\Lambda}$ composition-adjusted DERIVED Defined on: arms in the TCGA outcome panel (145) $\hat{\Lambda}$ TCGA legs only; the GSE leg is bare-stem matched
surv_os_q_adj ARM P-VALUE 6%	BH-adjusted q $\hat{\Lambda}$ composition-adjusted DERIVED Defined on: arms in the TCGA outcome panel (145) $\hat{\Lambda}$ TCGA legs only; the GSE leg is bare-stem matched

AID_ $\hat{\Lambda}$ 4

reference $\hat{\Lambda}$ context, provenance and rarely-quoted detail

Arm-IDENTIFIABILITY rollup for the arm: how often this arm is separable from its same-seed mates (arm_resolvable , arm_sep_z , oof_drho medians). 4 columns $\hat{\Lambda}$ coverage 6%$\hat{\Lambda}$6% (median 6%) $\hat{\Lambda}$ 3 share the block description 1x fraction 0$\hat{\Lambda}$1 $\hat{\Lambda}$ 1x real $\hat{\Lambda}$ 0 $\hat{\Lambda}$ 1x signed $\hat{\Lambda}$1$\hat{\Lambda}$1 $\hat{\Lambda}$ 1x count	
aid_frac_resolvable ARM FRACTION 0$\hat{\Lambda}$1 AGG ARM-IN-FAMILY 6%	fraction DERIVED Defined on: arms inside $\hat{\Lambda}$1 MULTI-ARM (gene,family) cell $\hat{\Lambda}$ the arm rung's 20.3%-of-edges footprint (142)
aid_med_arm_sep_z ARM REAL $\hat{\Lambda}$ 0 AGG ARM-IN-FAMILY 6%	median $\hat{\Lambda}$ per arm $\hat{\Lambda}$ z-score DERIVED Defined on: arms inside $\hat{\Lambda}$1 MULTI-ARM (gene,family) cell $\hat{\Lambda}$ the arm rung's 20.3%-of-edges footprint (142)
aid_med_oof_drho ARM SIGNED $\hat{\Lambda}$1$\hat{\Lambda}$1 AGG ARM-IN-FAMILY 6%	median $\hat{\Lambda}$ out-of-fold DERIVED Defined on: arms inside $\hat{\Lambda}$1 MULTI-ARM (gene,family) cell $\hat{\Lambda}$ the arm rung's 20.3%-of-edges footprint (142)
aid_n_cells ARM COUNT AGG ARM-IN-FAMILY 6%	Multi-arm cells this arm participates in $\hat{\Lambda}$ fill 5.8% (the arm identifiability block is the sparsest on the card). AUTHORED

TIER_ $\hat{\Lambda}$ 5

reference $\hat{\Lambda}$ context, provenance and rarely-quoted detail

Expression tier of the arm against the detection floor $\hat{\Lambda}$ class, fraction of samples expressed, p90 and max log2 RPM. 5 columns $\hat{\Lambda}$ coverage 91%$\hat{\Lambda}$91% (median 91%) $\hat{\Lambda}$ 2 share the block description 2x real $\hat{\Lambda}$ 0 $\hat{\Lambda}$ 1x categorical $\hat{\Lambda}$ 1x boolean $\hat{\Lambda}$ 1x fraction 0$\hat{\Lambda}$1	
tier_class ARM CATEGORICAL 91%	Expression tier $\hat{\Lambda}$ silent on 1,518 of 2,236 arms (67.9%), conditional 476, the rest robust. AUTHORED
tier_expressed ARM BOOLEAN 91%	Is the arm above the expression floor often enough to be usable (718 true / 1,518 false)? AUTHORED
tier_frac_expressed ARM FRACTION 0$\hat{\Lambda}$1 91%	fraction DERIVED Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)
tier_max ARM REAL $\hat{\Lambda}$ 0 91%	maximum DERIVED Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)

tier_p90

ARMREAL 91%

The arm's 90th-percentile RPM across tumour samples (0.10 – 18.49, median 0.94). AUTHORED

WSHIFT_ 6

reference context, provenance and rarely-quoted detail

Within-patient paired quantities on the ~103 matched tumour/NAT participants. 6 columns coverage 45%–91% (median 52%) 6 share the block description 3x real 2x real, signed 1x count		
wshift_mean_dGlobalRank	mean	DERIVED
ARMREAL SIGNED52%	Defined on: arms in the WITHIN-PATIENT paired subset (n% ^ 103 matched tumour/NAT pairs; 2,236 arms)	
wshift_mean_own_shift	mean shift	DERIVED
ARMREAL SIGNED52%	Defined on: arms in the WITHIN-PATIENT paired subset (n% ^ 103 matched tumour/NAT pairs; 2,236 arms)	
wshift_n_pairs	count	DERIVED
ARMCOUNT91%	Defined on: arms in the WITHIN-PATIENT paired subset (n% ^ 103 matched tumour/NAT pairs; 2,236 arms)	
wshift_nat_patient_sd	in matched normal (NAT) standard deviation	DERIVED
ARMREAL 74%	Defined on: arms in the WITHIN-PATIENT paired subset (n% ^ 103 matched tumour/NAT pairs; 2,236 arms)	
wshift_own_specific_frac	fraction	DERIVED
ARMREAL 45%	Defined on: arms in the WITHIN-PATIENT paired subset (n% ^ 103 matched tumour/NAT pairs; 2,236 arms)	
wshift_sd_dGlobalRank	standard deviation	DERIVED
ARMREAL 45%	Defined on: arms in the WITHIN-PATIENT paired subset (n% ^ 103 matched tumour/NAT pairs; 2,236 arms)	

SURROGATE_ 2

reference context, provenance and rarely-quoted detail

2 columns coverage 0%–0% (median 0%) 1x fraction 1x categorical		
surrogate_corr	Correlation between this arm and its surrogate (0.62 – 0.89) the surrogate's quality gate. See the edge twin. AUTHORED	
ARMFRACTION 0%1	Defined on: arms with a same-seed SURROGATE (MH-166) 3.6% of edges	
surrogate_instrument	Which surrogate stands in for this arm's collapsed GTEx leg (4 levels). AUTHORED	
ARMCATEGORICAL0%		

DISC_ 2

reference context, provenance and rarely-quoted detail

2 columns coverage 1%–100% (median 50%) 1x categorical 1x count		
--	--	--

disc_gold_families	Which seed families this row's discovery-queue edges belong to, semicolon-separated. Usually ONE at the arm rung (an arm has one seed). AUTHORED
ARM CATEGORICAL 1%	
disc_n_gold_edges	How many of the 157 discovery-queue edges name this arm. Concentrated: miR-106b-5p 39 Â· miR-20a-5p 25 Â· miR-93-5p 25 Â· miR-19a-3p 19 Â· miR-18a-5p 9. AUTHORED
ARM COUNT 100%	

FOC_ Â· 7
reference â€” context, provenance and rarely-quoted detail

Paralog-decontaminated FOCAL-locus concordance.

7 columns Â· coverage **30%â€”30%** (median 30%) Â· **5** share the block description
2x count Â· 2x signed â^'1â€|1 Â· 1x identifier Â· 1x boolean Â· 1x p-value

foc_focal_locus	The focal MIRBase locus used for this arm's CN analysis (437 distinct). AUTHORED
ARM IDENTIFIER 30%	

foc_multilocus	Does this arm map to MORE THAN ONE genomic locus (35 true / 698 false)? AUTHORED
ARM BOOLEAN 30%	

foc_n	count DERIVED
ARM COUNT 30%	<i>Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) â€” â” DESCRIPTIVE, not an instrument</i>

foc_n_loci	count DERIVED
ARM COUNT 30%	<i>Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) â€” â” DESCRIPTIVE, not an instrument</i>

foc_partial_p	p-value DERIVED
ARM P-VALUE 30%	<i>Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) â€” â” DESCRIPTIVE, not an instrument</i>

foc_partial_rho	Spearman Î DERIVED
ARM SIGNED Â^'1â€ 1 30%	<i>Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) â€” â” DESCRIPTIVE, not an instrument</i>

foc_spearman_rho	Spearman Î DERIVED
ARM SIGNED Â^'1â€ 1 30%	<i>Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) â€” â” DESCRIPTIVE, not an instrument</i>

CNV_ Â· 9
reference â€” context, provenance and rarely-quoted detail

Copy number at the arm's own locus â€” mean CN and amplification/deletion summaries.

9 columns Â· coverage **35%â€”35%** (median 35%) Â· **8** share the block description
6x percent Â· 3x real â‰¥ 0

cnv_mean_cn	mean DERIVED
ARM REAL Â‰¥ 0 35%	<i>Defined on: arms with a mapped CN locus in the cohort stratum (856) â€” â” multi-locus arms are DILUTED here</i>

cnv_median_cn	median DERIVED
ARM REAL Â‰¥ 0 35%	<i>Defined on: arms with a mapped CN locus in the cohort stratum (856) â€” â” multi-locus arms are DILUTED here</i>

cnv_pct_altered ARM PERCENT 35%	percent DERIVED Defined on: arms with a mapped CN locus in the cohort stratum (856) â€” multi-locus arms are DILUTED here
cnv_pct_amp ARM PERCENT 35%	percent DERIVED Defined on: arms with a mapped CN locus in the cohort stratum (856) â€” multi-locus arms are DILUTED here
cnv_pct_deep_del ARM PERCENT 35%	Percent of samples with a deep deletion at this arm's locus. AUTHORED
cnv_pct_gain ARM PERCENT 35%	percent DERIVED Defined on: arms with a mapped CN locus in the cohort stratum (856) â€” multi-locus arms are DILUTED here
cnv_pct_loh ARM PERCENT 35%	percent DERIVED Defined on: arms with a mapped CN locus in the cohort stratum (856) â€” multi-locus arms are DILUTED here
cnv_pct_loss ARM PERCENT 35%	percent DERIVED Defined on: arms with a mapped CN locus in the cohort stratum (856) â€” multi-locus arms are DILUTED here
cnv_sd_cn ARM REAL ≥ 0 35%	standard deviation DERIVED Defined on: arms with a mapped CN locus in the cohort stratum (856) â€” multi-locus arms are DILUTED here

CNVC_ 5

reference â€” context, provenance and rarely-quoted detail

CN-to-expression concordance fit for the arm (756 arms): does locus CN actually move this arm's abundance. 5 columns â€” coverage 31%â€”31% (median 31%) â€” 4 share the block description 2x p-value â€” 2x signed $\hat{\rho} \in [-1, 1]$ â€” 1x count

cnvc_n ARM COUNT 31%	count DERIVED Defined on: arms with a CNâ†”expression concordance fit (756)
cnvc_partial_p ARM P-VALUE 31%	p-value DERIVED Defined on: arms with a CNâ†”expression concordance fit (756)
cnvc_partial_q ARM P-VALUE 31%	BH q for the arm's CN-coupling partial correlation. AUTHORED
cnvc_partial_rho ARM SIGNED $\hat{\rho} \in [-1, 1]$ 31%	Spearman $\hat{\rho}$ DERIVED Defined on: arms with a CNâ†”expression concordance fit (756)
cnvc_spearman_rho ARM SIGNED $\hat{\rho} \in [-1, 1]$ 31%	Spearman $\hat{\rho}$ DERIVED Defined on: arms with a CNâ†”expression concordance fit (756)

MODEL_ 1

reference â€” context, provenance and rarely-quoted detail

1 columns â€” coverage 30%â€”30% (median 30%) 1x count
--

model_n_genes

Which model variant produced this row.

AUTHORED

ARM

COUNT

AGG EDGE

30%

| Defined on: arms carrying >=1 curated HE edge in the model design (741)

COMV_ 2

reference context, provenance and rarely-quoted detail

2 columns coverage 15%15% (median 15%)

1x count 1x identifier

comv_module

Which co-movement module this arm belongs to in the 6b-RETIRED heuristic lane.

AUTHORED

ARM

COUNT

15%

comv_top_partner

Co-expression module membership and top co-expressed partner. Pure co-expression the cross-state class columns are deliberately excluded from this block.

AUTHORED

ARM

IDENTIFIER

15%

| Defined on: arms in a co-expression module (371) pure co-expression; the state-class columns are excluded

CLOAD_ 3

reference context, provenance and rarely-quoted detail

Compartment LOADING of the arm.

3 columns coverage 8%60% (median 60%) 2 share the block description

1x real 0 1x fraction 01 1x categorical

cload_dose_rpm

dose

DERIVED

ARM

REAL 0

60%

| Defined on: arms with a compartment-loading row (1,473) its source has NO PRODUCER (MH-111 ad-hoc)

cload_lock

Is this arm's compartment loading LOCKED to one cell type rather than spread across the deconvolution?

AUTHORED

ARM

FRACTION 01

8%

cload_top_compartment

the top-ranked

DERIVED

ARM

CATEGORICAL

60%

| Defined on: arms with a compartment-loading row (1,473) its source has NO PRODUCER (MH-111 ad-hoc)

HX_ 1

reference context, provenance and rarely-quoted detail

1 columns coverage 28%28% (median 28%)

1x real 0

hx_hly_baseline_imputed

IMPUTED healthy baseline (miTED rank-transfer or seed-mate substitution) LABELLED as imputed, never presented as a measurement.

AUTHORED

ARM

REAL 0

28%

seed_family_card.tsv

seed_family

One row per the seed family itself, gene-free. 83.7% of families are SINGLETONS for those every 'family-level' statistic is trivially that one arm's value. Mask on fam_single_member first.

33 columns 1,959 rows 2 blocks median coverage 92%

Contents 2 blocks, most important first

1. seed_1 CORE
2. fam_32 REFERENCE

SEED_1

core the columns findings rest on

1 columns coverage 100% (median 100%)
1x identifier

seed_family

KEY the seed family itself, one row per family.

AUTHORED

KEY IDENTIFIER 100%

FAM_32

reference context, provenance and rarely-quoted detail

Seed-family structure the arm sits in membership, affinity and dominance within it.

32 columns coverage 4% (median 91%) 16 share the block description
12x count 8x real 7x fraction 1 2x boolean 1x categorical 1x identifier 1x signed
1

fam_ctx_composition

FAMILY CATEGORICAL 29%

The family's genomic-context MIX as a tally string sense_coding_host:1 on 251 families, sense_lncRNA_host:1 on 175, antisense_overlap:1 next (29 distinct patterns).

AUTHORED

fam_ctx_context_homogeneous

FAMILY BOOLEAN 29%

Do ALL of this family's members share one genomic context True on 524 of 571 (91.8%)?

AUTHORED

fam_ctx_dose_coding_host_frac

FAMILY FRACTION 1

29%

dose fraction DERIVED

Defined on: seed families in the arm-card universe (1,959; 83.7% single-member shares/HHI are 1 or NaN)

fam_ctx_dose_cotranscribed_frac

FAMILY FRACTION 1

29%

dose fraction DERIVED

Defined on: seed families in the arm-card universe (1,959; 83.7% single-member shares/HHI are 1 or NaN)

fam_ctx_n_distinct_coding_hosts

FAMILY COUNT 29%

count DERIVED

Defined on: seed families in the arm-card universe (1,959; 83.7% single-member shares/HHI are 1 or NaN)

fam_cur_degree_max FAMILY COUNT 36%	maximum DERIVED Defined on: seed families in the arm-card universe (1,959; â€™ 83.7% single-member â€™ shares/HHI are 1 or NaN)
fam_dominant_arm FAMILY IDENTIFIER 92%	the dominant one Â· per arm DERIVED Defined on: seed families in the arm-card universe (1,959; â€™ 83.7% single-member â€™ shares/HHI are 1 or NaN)
fam_dominant_share FAMILY FRACTION 0Â€"1 92%	The top member's share of the family's total dose. AUTHORED
fam_dose_hhi FAMILY FRACTION 0Â€"1 15%	Concentration of DOSE across the family's members. âœ… FLOOR-CORRECTED â€” verified 2026-08-19 : its minimum sits far below the raw 1/k floor at every member count (0.0000 at k=2 where the raw floor is 0.500), and spearman(n_members, hhi) = +0.1896 rather than the strongly negative value a raw index would give. AUTHORED
fam_dose_max_log2 FAMILY REAL Â%¥ 0 92%	dose Â· maximum DERIVED Defined on: seed families in the arm-card universe (1,959; â€™ 83.7% single-member â€™ shares/HHI are 1 or NaN)
fam_dose_med_log2 FAMILY REAL Â%¥ 0 92%	dose Â· median DERIVED Defined on: seed families in the arm-card universe (1,959; â€™ 83.7% single-member â€™ shares/HHI are 1 or NaN)
fam_dose_total_log2 FAMILY REAL Â%¥ 0 100%	dose Â· total DERIVED Defined on: seed families in the arm-card universe (1,959; â€™ 83.7% single-member â€™ shares/HHI are 1 or NaN)
fam_fame_npmid_sum FAMILY COUNT 100%	sum DERIVED Defined on: seed families in the arm-card universe (1,959; â€™ 83.7% single-member â€™ shares/HHI are 1 or NaN)
fam_n_expressed FAMILY COUNT 100%	count DERIVED Defined on: seed families in the arm-card universe (1,959; â€™ 83.7% single-member â€™ shares/HHI are 1 or NaN)
fam_n_guide FAMILY COUNT 100%	count DERIVED Defined on: seed families in the arm-card universe (1,959; â€™ 83.7% single-member â€™ shares/HHI are 1 or NaN)
fam_n_members FAMILY COUNT 100%	count Â· family members DERIVED Defined on: seed families in the arm-card universe (1,959; â€™ 83.7% single-member â€™ shares/HHI are 1 or NaN)
fam_n_members_context FAMILY COUNT 29%	count Â· family members DERIVED Defined on: seed families in the arm-card universe (1,959; â€™ 83.7% single-member â€™ shares/HHI are 1 or NaN)
fam_n_model_edges FAMILY COUNT 100%	count DERIVED Defined on: seed families in the arm-card universe (1,959; â€™ 83.7% single-member â€™ shares/HHI are 1 or NaN)
fam_n_passenger FAMILY COUNT 100%	count DERIVED Defined on: seed families in the arm-card universe (1,959; â€™ 83.7% single-member â€™ shares/HHI are 1 or NaN)
fam_n_shift_gt20_gated FAMILY COUNT 100%	count Â· shift Â· gated DERIVED Defined on: seed families in the arm-card universe (1,959; â€™ 83.7% single-member â€™ shares/HHI are 1 or NaN)

fam_real_n_c10_sum FAMILY COUNT 100%	count $\hat{\cdot}$ at the $\hat{\cdot}^{0.10}$ threshold $\hat{\cdot}$ sum DERIVED Defined on: seed families in the arm-card universe (1,959; $\hat{\cdot}$ 83.7% single-member $\hat{\cdot}^*$ shares/HHI are 1 or NaN)
fam_real_n_scored_sum FAMILY COUNT 100%	count $\hat{\cdot}$ sum DERIVED Defined on: seed families in the arm-card universe (1,959; $\hat{\cdot}$ 83.7% single-member $\hat{\cdot}^*$ shares/HHI are 1 or NaN)
fam_redund_m_eff FAMILY REAL $\hat{\cdot}^{\%} 0$ 4%	EFFECTIVE number of independent members after accounting for within-family correlation $\hat{\cdot}$ the family's redundancy-corrected size (mode 2.0 , 50 distinct values). AUTHORED
fam_redund_median_within_rho FAMILY SIGNED $\hat{\cdot}^{*1}\hat{\cdot} 1$ 4%	median $\hat{\cdot}$ Spearman $\hat{\cdot}$ DERIVED Defined on: seed families in the arm-card universe (1,959; $\hat{\cdot}$ 83.7% single-member $\hat{\cdot}^*$ shares/HHI are 1 or NaN)
fam_redund_overcount_frac FAMILY FRACTION $\hat{\cdot}^{\%} 1$ 4%	fraction DERIVED Defined on: seed families in the arm-card universe (1,959; $\hat{\cdot}$ 83.7% single-member $\hat{\cdot}^*$ shares/HHI are 1 or NaN)
fam_seed_shift_max_gated FAMILY FRACTION $\hat{\cdot}^{\%} 1$ 13%	shift $\hat{\cdot}$ maximum $\hat{\cdot}$ gated DERIVED Defined on: seed families in the arm-card universe (1,959; $\hat{\cdot}$ 83.7% single-member $\hat{\cdot}^*$ shares/HHI are 1 or NaN)
fam_single_member FAMILY BOOLEAN 100%	$\hat{\cdot}$ THE DOMINANT FACT ABOUT THIS ENTIRE CARD: 1,639 of 1,959 families (83.7%) are SINGLETONS. For those rows every 'family-level' statistic is TRIVIALY that one arm's value, and fam_dominant_share is forced to exactly 1.0. AUTHORED
fam_targetome_seed_med FAMILY REAL $\hat{\cdot}^{\%} 0$ 97%	median DERIVED Defined on: seed families in the arm-card universe (1,959; $\hat{\cdot}$ 83.7% single-member $\hat{\cdot}^*$ shares/HHI are 1 or NaN)
fam_targetome_seq_med FAMILY REAL $\hat{\cdot}^{\%} 0$ 30%	median DERIVED Defined on: seed families in the arm-card universe (1,959; $\hat{\cdot}$ 83.7% single-member $\hat{\cdot}^*$ shares/HHI are 1 or NaN)
fam_var_hhi FAMILY FRACTION $\hat{\cdot}^{\%} 1$ 14%	Herfindahl concentration of abund_sd ACROSS the family's members $\hat{\cdot}$ is the family's variance carried by one arm? AUTHORED
fam_var_max FAMILY REAL $\hat{\cdot}^{\%} 0$ 89%	maximum DERIVED Defined on: seed families in the arm-card universe (1,959; $\hat{\cdot}$ 83.7% single-member $\hat{\cdot}^*$ shares/HHI are 1 or NaN)
fam_var_med FAMILY REAL $\hat{\cdot}^{\%} 0$ 89%	median DERIVED Defined on: seed families in the arm-card universe (1,959; $\hat{\cdot}$ 83.7% single-member $\hat{\cdot}^*$ shares/HHI are 1 or NaN)